

Ecological Economics

Asset Stranding in the Agri-Food Sector: A Case for Divestment?

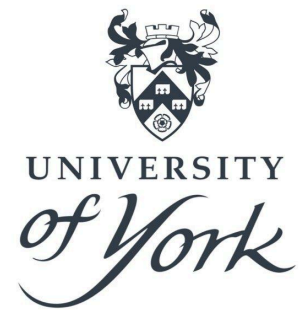
--Manuscript Draft--

Manuscript Number:	ECOLEC-D-26-02369
Article Type:	Analysis
Section/Category:	Analysis
Keywords:	divestment, asset stranding, financial risk, agri-food, asset pricing, efficient markets, climate finance
Corresponding Author:	Nikhil Ghosh University of York
First Author:	Nikhil Ghosh
Order of Authors:	Nikhil Ghosh
	Truzaar Dordi
	Julio Galindo-Gutiérrez
	Alain Naef
Abstract:	Asset stranding risks have motivated financial activists to advocate divestment in the agri-food sector, mirroring the long running fossil fuel divestment campaign. Given the dual physical and transition risks that the agri-food sector faces in plausible climate scenarios, financial institutions may have good reason to consider existing agri-food investments to maintain their positions. If such asset standing risks are priced, it should be expected that divested portfolios outperform the market. To test this logic, this paper presents the results of a sequential divestment experiment in the agri-food sector from 2016-2026 in the USA and Europe. We find that divestment has little impact on risk adjusted returns. Additionally, the return spread on a "greener" and "browner" portfolio is minimal and insignificant in explaining portfolio returns in either market. Some heterogenous results are present between the two markets, which may affect financial institutions' decision to divest. We interpret these results as a consequence of the adaptive nature of agri-food equity markets in response to climate risk, and suggest that public policy efforts are necessary to make divestment campaigns financially.
Opposed Reviewers:	

Nikhil Feroze Ghosh

Department of Environment and Geography
University of York
York
YO10 5DD
United Kingdom

nikhil.ghosh@york.ac.uk



Dear Prof. Dr. Stefan Baumgärtner and Prof. Begüm Özkaynak,

I am pleased to submit for consideration to *Ecological Economics* the manuscript entitled "**Asset Stranding in the Agri-food Sector: A Case for Divestment?**". The paper is co-authored with Dr. Truzaar Dordi and Dr. Julio Galindo-Gutiérrez of the University of York, UK, and Dr. Alain Naef of ESSEC Business School, France. I confirm that this work is original, has not been published elsewhere, and is not currently under consideration for publication.

This research is the first to investigate the financial consequences of portfolio divestment in the agri-food sector. Specifically, we test the logic that portfolio divestment is a financially prudent strategy for investors looking to insulate themselves from the many stranded asset risks that the agri-food sector faces. Our primary finding is that divestment has little impact on portfolio performance, indicating that these risks are not currently being priced into agri-food equities against the market. Theoretically, our results lend credence to an adaptive, rather than efficient view of financial markets in the context of environmental risk, suggesting that there are structural barriers to the pricing of this risk in addition to the typical information problems suggested by the efficient markets literature.

In line with the aims and objectives of *Ecological Economics*, our work studies the relationship between the biophysical pressures that the food system faces and the economics of the financial system. Readers of the journal will be interested in three novel aspects of our paper. First, there has been scant empirical analysis of the economics of asset stranding risk in the agri-food sector, a theme on which our paper opens a broad academic dialogue. Second, our work is methodologically innovative by combining the absolute divestment approach with recent advances in the use of "green factors" in explaining financial portfolio returns. Finally, our work advances theoretical perspectives on financial markets, suggesting that attention to how markets adapt to changes in institutional structures within the financial system are of key importance in allocating finance towards sustainable ends.

We hope this paper will be of interest to *Ecological Economics*, and thank you for your consideration.

Sincerely,

Mr. Nikhil Feroze Ghosh (corresponding author)

University of York
York YO10 5DD
york.ac.uk

Title: Asset Stranding in the Agri-Food Sector: A Case for Divestment?

Author(s): **Nikhil Feroze Ghosh^a (Corresponding Author)**, Truzaar Dordi^a, Julio Galindo-Gutiérrez^a, Alain Naef^b

a. Department of Environment and Geography, University of York, York UK

b. ESSEC Business School, Paris, FR

Correspondence Email: nikhil.ghosh@york.ac.uk

Correspondence Address:

Department of Environment and Geography

University of York

York

YO10 5DD

United Kingdom

Abstract

Asset stranding risks have motivated financial activists to advocate divestment in the agri-food sector, mirroring the long running fossil fuel divestment campaign. Given the dual physical and transition risks that the agri-food sector faces in plausible climate scenarios, financial institutions may have good reason to consider existing agri-food investments to maintain their positions. If such asset standing risks are priced, it should be expected that divested portfolios outperform the market. To test this logic, this paper presents the results of a sequential divestment experiment in the agri-food sector from 2016-2026 in the USA and Europe. We find that divestment has little impact on risk adjusted returns. Additionally, the return spread on a “greener” and “browner” portfolio is minimal and insignificant in explaining portfolio returns in either market. Some heterogenous results are present between the two markets, which may affect financial institutions' decision to divest. We interpret these results as a consequence of the adaptive nature of agri-food equity markets in response to climate risk, and suggest that public policy efforts are necessary to make divestment campaigns financially.

Keywords

divestment, asset stranding, financial risk, agri-food, asset pricing, efficient markets, climate finance

- First study to investigate the impacts of portfolio divestment in the agri-food sector
- Divested portfolios do not outperform the market over a 10 year period
- The return spread between greener and browner agri-food portfolios is negligible
- No evidence of asset stranding risk being priced into agri-food equities

Asset Stranding in the Agri-Food Sector: A Case for Divestment?

Abstract

Asset stranding risks have motivated financial activists to advocate divestment in the agri-food sector, mirroring the long running fossil fuel divestment campaign. Given the dual physical and transition risks that the agri-food sector faces in plausible climate scenarios, financial institutions may have good reason to consider existing agri-food investments to maintain their positions. If such asset standing risks are priced, it should be expected that divested portfolios outperform the market. To test this logic, this paper presents the results of a sequential divestment experiment in the agri-food sector from 2016-2026 in the USA and Europe. We find that divestment has little impact on risk adjusted returns. Additionally, the return spread on a “greener” and “brownier” portfolio is minimal and insignificant in explaining portfolio returns in either market. Some heterogenous results are present between the two markets, which may affect financial institutions’ decision to divest. We interpret these results as a consequence of the adaptive nature of agri-food equity markets in response to climate risk, and suggest that public policy efforts are necessary to make divestment campaigns financially.

1 Introduction

The global agri-food system contributes around a quarter of global greenhouse gas emissions and is a leading driver of deforestation, the introduction of novel pollutants and harmful nutrient flows, and biodiversity loss (Crippa et al., 2021; Intergovernmental Panel on Climate Change, n.d.; te Wierik et al., 2025). Equally, agri-food is the sector perhaps most exposed to climate risks, with yield reduction and soil quality degradation being one of the most significant hazards the world economy faces under likely climate scenarios (Jones et al., 2026; Shukla et al., 2019; Vermeulen et al., 2012). These features make the agri-food sector uniquely exposed to the risk of asset stranding under different sustainability transition pathways. Asset stranding refers to the pre-mature write down or de-valuation of physical or financial assets

1
2
3
4 in the event of an industrial transition (Caldecott, 2018), and is increasingly understood as
5 one of the most significant mechanisms determining the flow of financial resources towards
6 sustainable and unsustainable forms of production and consumption (Mehrabi, 2026; Walton
7 et al., 2025).

8
9
10 In this paper, we analyse the specificities of asset stranding in the agri-food sector and
11 discuss their implications for divestment as a technique of sustainable corporate governance
12 by institutional investors and financial activists. Asset stranding risks are made material
13 for investors as the write-downs of productive assets feed through to the value of financial
14 assets such as debt and equity, the holders of which may be under-compensated. As climate
15 change comes to bear on corporate revenues, or as corporations retire such unsustainable
16 assets voluntarily or by regulatory force, any negative effects on returns may leave investors
17 in the lurch. Investors may therefore wish to divest early in order to reduce their exposure
18 to potentially worthless assets in the future. In the energy industry, fossil fuel assets have
19 long displayed these characteristics, motivating much public and academic debate over the
20 effectiveness of fossil fuel divestment by institutional investors as a mechanism of sustainable
21 corporate governance (Braungardt et al., 2019; Dordi et al., 2025). Divestment is often
22 justified using this precise asset stranding logic—investors should get out early to maintain
23 their position during a sustainability transition.

24
25 If asset stranding risks were being priced, we would expect divestment to affect portfolio
26 returns (Ansar et al., 2013; Braungardt et al., 2019). Whether or not this is the case is
27 important, since there have been recent calls by financial activists for institutional investors to
28 evaluate their investments in agri-food corporations, and possibly divest if they consider them
29 to carry asset stranding risk (Friends of the Earth International, 2026; GlobalCanopy.org, 2026;
30 WWF, 2022). Research is therefore needed on the impacts and effectiveness of divestment as
31 a corporate governance tool for sustainable transitions in the agri-food sector. This paper
32 advances this agenda. We pose the following three research questions:

- 33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65
- To what extent does the risk-return profile of the agri-food sector vary compared to the market as a whole?
 - How does divestment from the agri-food sector affect portfolio returns when compared to a relevant benchmark?
 - Is the spread between “greener” and “brownier” agri-food portfolios significant in explaining portfolio returns?

We answer the first two questions by constructing a set of agri-food specific and agri-food free equity portfolios and comparing their risk-reward characteristics in the USA and Europe

1
2
3
4 respectively using a Fama–French Five Factor Model. For the third question, we use firm
5 level data on Scope 3 emissions intensity as a proxy for stranded asset risk, we estimate a
6 “Green-Minus-Brown” spread for the agri-food sector as a whole. Ours is therefore the first
7 study to apply the divestment methodology to the agri-food sector. In doing so, our study
8 contributes to the rapidly emerging debates on whether capital markets are pricing asset
9 stranding risk in sectors beyond energy, and further offer a practical framework for financial
10 institutions and activists assessing asset stranding risk in the food sector.

11
12 Our primary results are as follows. First, divestment from segments of the agri-food
13 sector exposed to asset stranding risk does not reduce returns in US and European markets
14 when risk is accounted for. Consequently, investors concerned with the environmental or
15 social performance of agri-food corporations can do so without much detriment. We interpret
16 this result as resulting from a lack of environmental transition risk pricing in the agri-food
17 sector. We further support this interpretation by finding a small but negligible positive
18 return spread between a best and worst in class portfolio by Scope 3 emissions intensity,
19 suggesting investors are not penalising environmentally underperforming agri-food companies
20 in either market. Overall, our results suggest that any pricing of asset stranding risk by
21 investors in the agri-food sector have yet to show any medium to long term effect on asset
22 prices. We locate the reason for this in two primary issues. First, the structural conditions
23 for sustainability transitions in the agri-food sector are not in place—in particular the lack
24 of market facing alternatives to the existing food system and public policy support for
25 industrial transitions. This constrains the possibility of divestment and re-investment by
26 large financial institutions. Second, there is a lack of sector specific high quality firm level
27 data on environmental performance beyond greenhouse gas emissions to assess sustainability
28 transitions in the agri-food sector such as biodiversity impact, deforestation, soil degradation,
29 and chemical flows (Caldecott et al., 2021). The lack of this data makes it challenging for
30 investors to assess the environmental performance of agri-food corporations and price their
31 risks into assets. In other words, there is no clear relationship between asset stranding risks
32 in future climate scenarios and the present performance of agri-food equities.

33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 **2 Literature Review**

52
53 In this section we review the conceptual and empirical literature on asset stranding and
54 divestment in sustainability transition contexts. Subsequently, we explore the various kinds
55 of stranded asset risks in the agri-food sector under plausible sustainable food transition
56 scenarios based on academic and policy literatures, forming the basis for our empirical
57 analysis.

2.1 Asset Stranding Risk and the Agri-food Sector

Asset stranding is increasingly understood as a crucial mechanism for enabling or constraining sustainability transitions, as it influences the decision of institutional investors to finance unsustainable companies (Caldecott, 2017). Asset stranding denotes a situation in which the economic value of an asset in the real economy must be written down in climate scenarios in which that asset is made obsolete (Caldecott et al., 2021; van der Ploeg & Rezai, 2020). This has the effect of reducing the value of financial assets, leading to potentially significant losses for investors.

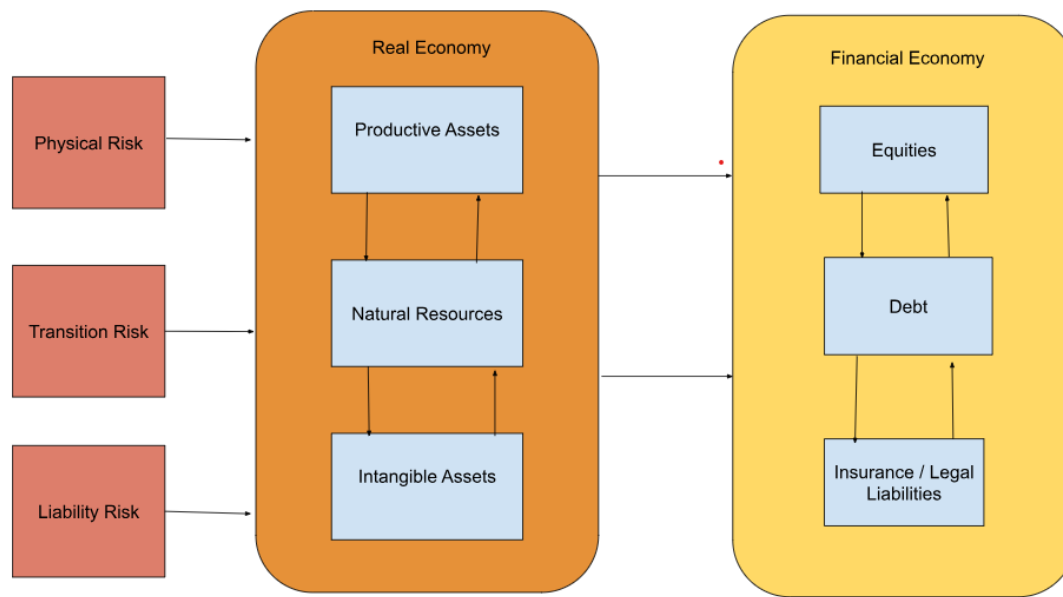


Figure 1. Transmission of Climate Risks to Financial Assets.

Source: Author's elaboration based on Caldecott et al. (2021) and van der Ploeg and Rezai (2020).

Figure 1 shows the chain of transmission from a climate-related risk through assets in the real economy to those in the financial economy. Climate risks are understood using the framework proposed by Carney (2015) as comprising physical risks from direct damages, transition risks from assets rapidly de-valued during industrial transitions, and liability risks resulting from claims for damages from parties. Through different channels, these risks affect potentially stranded tangible assets in the agri-food sector include production, processing, storage, and distribution facilities, real assets which stand to lose value in the event of climate related food crises such as droughts or floods, or in the event of transitions to sustainable food (Mehrabi, 2026; Walton et al., 2025). Additionally, it includes intangible assets such as intellectual property and branding, natural resources such as land, raw minerals, or water

bodies that a given corporation may own. Financial assets are those which derive their fundamental value from the continued use of these assets in the real economy. There is now a wide literature exploring the effects of different dimensions of climate risk on many of the types of physical and financial assets existing in the economy (Giglio et al., 2021). Given the emphasis in this paper on equity divestment, we focus only on the transmission of two forms of climate risk—physical and transition—on productive assets which, in turn, affect the value of equities.

2.1.1 Physical Risks

The agri-food sector is threatened more than any other by the physical impacts of climate change and other nature related ecological changes. Indeed, advances in Anthropocene Science have shown that the feasibility of present cropping patterns is premised on the stable climatic and ecological conditions of the Holocene (Angus, 2016; Rockström et al., 2020). Plausible estimates show that in high and moderate emissions scenarios, global crop yields are estimated as reducing by an average of 7.8% in 2050 due to climate related extreme weather (Hultgren et al., 2025). One off, unforeseeable, extreme weather events cause the variance around the average to be large, with other studies showing annual losses up to 30% in recent extreme weather scenarios (Lesk et al., 2022). Further, there is a growing scientific understanding of the dependence of food systems on currently degrading features of natural ecosystems such as biodiversity, soil health, and groundwater quality (Dasgupta, 2021; IPBES, 2019; Jones et al., 2026; Webb et al., 2020).

For agri-food corporations across the value chain, these physical damages risk the under-utilisation of input and seed producing, processing, and distribution infrastructure. Consider an incidence of prolonged wheat crop failure due to extreme weather conditions (Gaupp et al., 2019; Gourdji et al., 2025; Lesk et al., 2022). This scenario would likely see a swift drop in demand for seeds and agro-chemicals, causing some production facilities to cease revenue generation. Grain milling and processing facilities would, in this case, remain underutilized and potentially have their value written down. Consequently, the equities of such companies stand to fall in value. Though agri-food corporations do have considerable adaptive capacity in such scenarios, it is increasingly plausible that, given the failure to mitigate global warming to 1.5 degrees above pre-industrial levels, there are limits to how much they can insulate the value of their productive assets from such crop failures. In sum, this scenario presents an argument for financial institutions to divest from companies exposed to such risk.

2.1.2 Transition Risks

The agri-food sector is exposed to a variety of transition risks. A uniting theme in many prominent recommendations for sustainable food systems is “agro-ecology” referring to the integration of ecological science with agriculture and foodways (Ewert et al., 2023). In practice, this refers to a wide range of food production methods which to a greater or lesser degree use fewer inorganic inputs, less soil disturbance, increase crop diversity and cover crop use, promote organic soil health and biodiversity, reduce food processing, and localise food supply chains (Gliessman, 2021). Another prominent shift focuses in particular on the reduced consumption of red meat and poultry, in particular from intensively farmed sources (Harwatt et al., 2020; Searchinger et al., 2019; Willett et al., 2019). This shift is motivated both by the excessive greenhouse gas emissions from intensive meat farming as well as the pressure it places on standing forests. In addition, the consumption of fish and other marine species is suggested to be reduced in particular from large trawling vessels due to their impact on marine biodiversity and stocks. Additional shifts include moving away from diets heavy on ultra-processed foods (Kesse-Guyot et al., 2023; Seferidi et al., 2020). All of these transitions, we posit, requires the decommissioning of large productive infrastructures, such as large agro-chemical plants in the case of agricultural inputs or various types of primary processing and manufacturing facilities in the case of food processing companies. Given the growing interest in sustainable food transitions it is plausible that such industrial transitions will be implemented by public and private actors.

Consider, for example, the case of a dietary transition involving substitution of red meat for legumes as a primary protein source in high income countries as recommended by the EAT-Lancet Commission on Planetary Health (Rockström et al., 2025). Such a transition may be spurred by the need to abate the high levels of greenhouse gas, in particular methane emissions—around 15% of global CO₂e emissions—contributed by livestock production (Crippa et al., 2021; Mandouri et al., 2025). With regard to deforestation there have been recent private and regulatory efforts to improve the scope and coverage of deforestation-free supply chains (Lambin & Furumo, 2023). In the EU, for instance, legal obligations for Deforestation Free Conversion and Sourcing have come into operation from the end of 2026, and it is likely that more stringent requirements will be implemented in the future (European Union, 2025). Meat processing and distribution corporations too have much to lose in this scenario. The economic value of the animal feeding, processing, and packing facilities of major meat companies such as Concentrated Animal Feeding Operations (CAFOs) or abattoirs stand to lose considerable value. Walton et al. (2025) estimate the value of such facilities at USD 227 billion for the 11 largest meat processing companies. Compounding the risks posed to physical infrastructure are liability risks, with an emerging number of legal cases against

the “Methane Majors”—largely animal agribusinesses—for their climate impact (Bray & Poston, 2024).

2.2 Asset Stranding Risk and Divestment

Asset stranded risks are widespread in the agri-food sector, and this provides a basis for investors to consider divestment in order to insulate themselves from future losses. Indeed, while there has not been to date large scale divestment campaigns in the agri-food system analogous to those in the fossil fuel industry there is growing interest from financial institutions to engage in environmental corporate governance (FSDA, 2024; Greenbank, 2024; Norges Bank Investment Management, 2021). Campaigns from civil society organisations around sustainable food production have catalysed much of this movement (Forest500, 2024; Friends of the Earth International, 2026; Global Witness, 2025; Midgeley et al., 2021; Portfolio Earth, 2021; S3F, 2026), identifying many of the most significant bad actors in the food sector. Divestment from these companies is an important weapon of choice (Friends of the Earth International, 2026; GlobalCanopy.org, 2026) however it is important that these efforts be backed by a clear understanding of the impacts and challenges associated with such a strategy. In this section, we therefore review the theoretical rationale for asset stranding risk motivating divestment.

For a given corporation, positive asset stranding risk will exert downward pressure on the value of the firm’s financial assets (Caldecott, 2018). Investors may demand a risk premium for continuing to hold assets exposed to climate related risks, or divest altogether should they believe that a firm is not sufficiently allocating resources to address that risk. However, there are different theoretical models of financial market behaviour which explain the (lack of) transmission of asset stranding risk from the real to the financial economy, and the role of asset prices in co-ordinating the transition of this risk. The connection between asset stranding risk and asset price is important to emphasise as it determines the degree to which unregulated capital markets can smoothly govern the allocation of capital away from unsustainable industries (cf. Ameli et al., 2020; Edmans, 2024).

In the efficient markets tradition of financial economics following Fama (1970), the extent to which investors are able to accurately price new information on particular climate risk factors is the key transmission channel (Alessi et al., 2021; Bolton & Kacperczyk, 2021). When new information regarding the climate risk of a negatively exposed security emerges—for example, the Paris Agreement in 2015—investors are able to incorporate these into their expectations regarding future cash flow from that security and revise prices downward (Campiglio et al., 2023). Consequently, the disclosure of climate related risks, either in

1
2
3
4 company level environmental reporting or media reporting about climate impacts (Ardia
5 et al., 2023), can serve as sufficient signals for markets to re-price assets such that they find
6 an equilibrium which maximises investor utility (Pástor et al., 2021). Empirically, the effects
7 of climate risks on asset prices can therefore be ascertained by proxying them as additional
8 risk factors in standard asset pricing models (Bolton & Kacperczyk, 2021, 2023; Pástor et al.,
9 2022). Failure to do so can be understood as an asymmetry in information transmission
10 from climate risk to financial risk (Hong et al., 2000, 2019), which may be improved by more
11 accurate risk metrics.
12

13
14 An alternative, more general, approach to interpreting the pricing of asset stranding risk
15 emerges from the evolutionary economic tradition which views financial markets as adaptive
16 rather than efficient (Farmer & Lo, 1999; Lo & Zhang, 2024; Silver, 2018). This view puts
17 far more emphasis on the *environment* in which investment decisions are made than the
18 asymmetries of information between market participants. Market efficiency is therefore better
19 understood as a “steady-state limit of a market in an unchanging financial environment” (Lo,
20 2017, p. 221). That financial markets in future climate scenarios are likely to be characterised
21 by institutional and structural change imply that an adaptive approach is a more prudent
22 approach to analysing asset prices in the context of the ecological crisis (Ameli et al., 2020;
23 Hall et al., 2017). Moreover, future climate scenarios are in general associated with high
24 degrees of uncertainty and non-linearity (Ackerman, 2018; Monasterolo, 2020), making the
25 typical analysis of portfolio risk as reflected in historical price volatility highly contingent.
26 Indeed, survey data indicates that finance professionals are more likely to believe that climate
27 risks are mispriced than academics due to the latter’s belief in market efficiency (Stroebe &
28 Wurgler, 2021).
29

30
31 Fossil fuel divestment is an important case study in the financial economics of asset
32 stranding risk. The divestment agenda has been taken up by a number of institutional
33 investors, often motivated by the logic that it is more financially prudent to exclude a sector
34 at increasing risk from asset stranding (Ayling & Gunningham, 2017; Dordi et al., 2025;
35 Trouvé & Naef, 2025). The empirical evidence on whether investors were rewarded by such
36 strategies is mixed. Studies such as Trinks et al. (2018) and Plantinga and Scholtens (2021)
37 find no impact on portfolio returns from fossil fuel divestment in a USA and Global sample
38 respectively over a long time period. Hunt and Weber (2019) find in the Canadian context,
39 over a short period of time, that divestment slightly improves risk adjusted returns and
40 decreases the carbon intensity of portfolios. In a wider review, Plantinga and Scholtens (2024)
41 find that there is generally little impact on performance over a long time frame.
42

43
44 If these results are viewed through an efficient markets lens, widespread public demands
45 for divestment from fossil fuels may appear as a storm in a teacup—the small or non-
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65

existent effects on returns of the fossil-fuel free portfolios to imply that investors are already pricing asset stranding risk into equity prices. Corporate carbon disclosures or ESG ratings may be serving as sufficient signals (Bolton & Kacperczyk, 2021). An adaptive market orientation suggests that structural and behavioral issues may be countervailing in the smooth transmission of climate related risk to asset prices (Silver, 2018). In effect, the lack of any return anomaly suggests the existence of a “carbon bubble”—overconfidence on the part of investors on the extent of write-downs that will come from decarbonization of energy. In an adaptive framework, investors may simply not believe that the threat of transitions are sufficient to compromise asset returns, as past prices give them little historical basis for doing so (Ameli et al., 2020; Silver, 2018; Thomä & Chenet, 2017).

The mixed evidence motivates research applying the divestment logic to other sectors. Three features of the agri-food sector makes a particularly interesting case study to do so for three primary reasons. First, as noted above, it is exposed to considerable physical risk in addition to the typical transition risk which motivates divestment in the fossil fuel industry. It is conceivable, for instance, that an investor may wish to exit investment from a grain commodity trader, on the assumption that such a firm might lose considerable value in the event of crop failures. Second, the agri-food sector is much different than that of the energy industry in that its risk profile has unique characteristics. Primary contributions to environmental harm from the food sector are not direct CO₂ emissions but rather categories such as methane emissions, deforestation, freshwater use, pollutants, and biochemical flows from the over-use of agricultural inputs which are non-standard in typical corporate disclosures. It is therefore an important sectoral case in the financial economics of environmental risk *outside* the most commonly studied category of direct CO₂ emissions. Finally, industrial processes in the agri-food sector which must change have a high degree of uncertainty in climate scenarios, as clear alternative technological options are missing. This is in contradistinction with the energy sector, where a relatively clear delineation is typically made between the sunset industry of fossil fuels and the sunrise industry of renewable energy (Semieniuk et al., 2021). Consequently, the “other side of the coin” in a typical divestment exercise is missing—if investors divest from harmful incumbent food companies, there are few clear options in which to re-invest. Given these characteristics, we suggest that the typical features of firms that investors use to price climate risk in other sectors, such as energy, are lacking. Consequently, divestment may produce different results.

3 Methods and Data

Since the pioneering work of Markowitz (1952), quantitative finance research typically uses a mean–variance framework to evaluate portfolio returns, comparing any anomalous returns between a target portfolio and that of a benchmark (Cochrane, 1994). From the standpoint of an investor aiming to maximize returns for a given level of risk, this comparison is what may motivate investment in green securities, all else being equal.

We follow the methodological approach taken in fossil fuel divestment research (Hunt et al., 2018; Trinks et al., 2018). This methodology involves constructing a number of alternative portfolios with different degrees of divestment from a given set of stocks exposed to risk and comparing them in terms of their risk-reward characteristics in a mean–variance framework. Practically, this represents an extreme case of exit based corporate governance in which investors exclude sub-classifications of the agri-food sector on principle. To do this, we constructed a suite of agri-food free portfolios from a given target universe, simulating a principles based divestment campaign. In order to tease out potentially heterogeneous results between geographical regions, we analysed two investable universes—the S&P 500, representing the USA, and the STOXX 600, representing Europe.

Using the Thomson Reuters Business Classification (TRBC) system, we first identified those codes which covered all business activities of interest in the agri-food system.¹ We categorised these into three coarse industry groups—Inputs, Processing, and Retail. Following Trinks et al. (2018), we then constructed a set of agri-food-free and sector focussed agri-food exclusive portfolios for both markets. We additionally created a composite portfolio of all agri-food stocks in the sample. Table 1 lists all of the portfolios analysed in the study along with their descriptions. Our study period ran from June 2016 to January 2026 (116 months), a period coinciding with increased global attention to the climate crisis following the signing of the Paris Agreement in April 2016 (Falkner, 2016; Huang & Zhai, 2021). It is therefore this period in which investors are likely to have assessed the potential for asset stranding and priced these into equities (Monasterolo & de Angelis, 2020; Silver, 2018).

We obtained monthly price and balance sheet data for each investable universe from the LSEG Worldscope database, in US Dollars. For each month in our study period, we first identified a list of all stocks listed in each index on the first date of a given month. This approach allowed us to include de-listed and dead stocks in our sample, eliminating survivorship bias (Gilbert & Strugnell, 2010). We then constructed each portfolio as a value weighted portfolio of all stocks, rebalanced each month to account for entry and exit, and

¹We selected the TRBC classification system over other commonly used systems such as the Global Industrial Classification System (GICS) or Standard Industrial Classification (SIC) codes because of its superior coverage of de-listed stocks on the database as well as its consistent classifications across countries.

Table 1. List of Portfolios Analysed. Portfolio descriptions denote the TRBC sub-sector classification codes included in the portfolio.

Portfolio Name	Description	# of Stocks (incl. de-listed)
Market_US	All listed stocks on the S&P 500	692
Market_EU	All listed stocks on the STOXX 600	931
<i>Agri-food Portfolios</i>		
Inputs_US	51101020 (Agricultural Chemicals) + 5210202015 (Agricultural Machinery) in Market_US	6
Inputs_EU	51101020 (Agricultural Chemicals) + 5210202015 (Agricultural Machinery) in Market_EU	5
Processing_US	54102010 (Fishing and Farming) + 54102020 (Food Processing) + 54101030 (Non-alcoholic Beverages) in Market_US	20
Processing_EU	54102010 (Fishing and Farming) + 54102020 (Food Processing) + 54101030 (Non-alcoholic Beverages) in Market_EU	21
Retail_US	54301020 (Food Retail and Distribution) + 53301020 (Restaurants & Bars) in Market_US	10
Retail_EU	54301020 (Food Retail and Distribution) + 53301020 (Restaurants & Bars) in Market_EU	23
All Agri-Food_US	Inputs_US + Processing_US + Retail_US	36
All Agri-Food_EU	Inputs_EU + Processing_EU + Retail_EU	49
<i>Divested Portfolios</i>		
Inputs_US_Div	Market_US excl. Inputs_US	686
Inputs_EU_Div	Market_EU excl. Inputs_EU	926
Processing_US_Div	Market_US excl. Processing_US	672
Processing_EU_Div	Market_EU excl. Processing_EU	936
Retail_US_Div	Market_US excl. Retail_US	682
Retail_EU_Div	Market_EU excl. Retail_EU	900
All_Agri-food_US_Div	Market_US excl. All_Agri-food_US	656
All_Agri-food_EU_Div	Market_EU excl. All_Agri-food_EU	882

estimated the returns of each. Risk adjusted returns were calculated using the Sharpe ratio (Sharpe, 1966) and Sortino ratio (Sortino & van der Meer, 1991). The Sharpe ratio weights a total portfolio return in excess of the risk free rate by its standard deviation. For a portfolio p ,

$$Sharpe_p = \frac{\mathbb{E}(R_p - R_f)}{SD(R_p)},$$

where R_p is the total portfolio return and R_f is the risk free rate.

To test for the statistical significance of differences in Sharpe ratios between portfolios, we utilise the Jobson–Korkie (JK) test (Jobson & Korkie, 1981) as well as the robust Ledoit–Wolf (LW) bootstrapped test with 5000 resamples (Ledoit & Wolf, 2008).

The Sortino ratio weights excess returns by a measure of downside risk, i.e. the minimum volatility that an investor would be willing to hold. Following similar studies in the literature (Lobe & Walkshäusl, 2016; Trinks et al., 2018), we use the risk free rate as the minimum measure of downside risk. For a portfolio p ,

$$Sortino_p = \frac{E(R_p - R_f)}{\sqrt{\frac{1}{T} \sum_{t=1}^T (Min(0, R_{p,t} - R_f))^2}}$$

We then compared the returns difference between the best-in-class and worst-in-class portfolios using their Scope 3 emissions intensity. Following the intuition of Bolton and Kacperczyk (2021), we expect that if investors were pricing asset stranding risk, firms with a higher emissions intensity may carry a “carbon premium”. Theoretically, Scope 3 intensity serves as a proxy for carbon related transition risks in the agri-food sector—firms with a higher Scope 3 intensity hold more physical assets at risk of stranding in transitions to low-carbon food production as proposed by the FAO (2023) and UNFCCC (2025) (see Figure 1). The focus on Scope 3 emissions rather than Scope 1 and 2 follows the unique carbon profile of the agri-food industry, in which the majority of emissions are associated with food production rather than direct and indirect energy use. Data for Scope 3 emissions was collected from the Carbon Disclosure Project (CDP) database. Following Hoffmann and Busch (2008), for stock i in month t , emissions intensity is calculated as

$$\text{Emissions Intensity}_{it} = \frac{\text{Reported Scope 3 emissions}_{it}}{\text{Company Revenue}_{it}}. \quad (1)$$

We further assess the role of environmental performance in asset pricing by constructing a Green Minus Brown (GMB) factor (Pástor et al., 2022), subtracting the returns of the best and worst performers for the companies in our composite portfolios AFC_US and AFC_EU.²

²We focus on this measure over others for reasons of data availability, i.e. data on Scope 3 emissions are

Using Scope 3 emissions data, we filtered out missing data and then sorted the stocks in our portfolio by emissions intensity for each month. We then constructed the best-in-class and worst-in-class portfolio as the respective value-weighted portfolios of the top and bottom quartile of emitters, and compared their historical returns. This approach gives us an analysis to conduct a *within sector* analysis of one measure of portfolio “greenness”, accounting for missing values and changes to reporting practices and frequencies.³ We therefore calculate the GMB spread in month t as

$$\text{GMB}_t = R_{\text{best},t} - R_{\text{worst},t}, \quad (2)$$

where R_{best} is the value weighted portfolio of the top quartile of emitters and R_{worst} is the value weighted portfolio of the bottom quartile.

Finally, we explain any anomalous portfolio returns using the Fama–French Five Factor model (Fama & French, 2016) both with and without the GMB factor as an additional control. The Five Factor model is an extension of the CAPM, controlling for four of the most significant firm level characteristics—firm size, book-to-market ratio, profitability, and investment. For a portfolio p in month t ,

$$\begin{aligned} R_{pt} - R_{ft} = & \alpha_p + \beta_0(R_{\text{Mkt},t} - R_{f,t}) + \beta_1 \text{SMB}_t + \beta_2 \text{HML}_t \\ & + \beta_3 \text{RMW}_t + \beta_4 \text{CMA}_t + \varepsilon_{pt}, \end{aligned} \quad (3)$$

where R_f is the risk free rate, R_{Mkt} is the return on the benchmark market portfolio, SMB is the return difference between a portfolio long on small cap and short on large cap stocks, HML is the return difference on a portfolio long on high book-to-market and short on low book-to-market stocks, RMW is the return difference on a portfolio long on high profitability and short on low profitability stocks, and CMA is the return difference on a portfolio long on low investment and short on high investment stocks. We obtained the factor values and risk-free rates for both markets from Kenneth French’s website⁴ and asset returns from Worldscope.

increasingly available at the firm level. In the agri-food sector, there are many other important domains of transition risk such as deforestation or soil health, however, data on firm level exposure to these is largely absent. Initiatives such as the Taskforce on Nature-Related Financial Disclosures (TNFD) aim to create standardised methodologies to measure these aspects, and as such data becomes available, there is scope for further research on their role as significant risk factors.

³We assume that in all periods, all firms in the sample use the same reporting procedures and are equally likely to report emissions. This assumption may be inaccurate, for instance, as large firms are more likely to report emissions than small ones. Moreover, emissions reporting is typically done annually. However, the data shows that the year on year difference in emissions is small, indicating that firms do not vary significantly in their emissions profile over a one year period.

⁴https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

4 Results

In this section we present the results of analysis. We first show the performance of agri-food portfolios and explain them using a Fama–French Five Factor model. We then show the results of a similar analysis of the inverse portfolios—portfolios which divest from the agri-food sector. We then provide an estimate and analysis of a “Green Factor” and integrate it as an additional control in the Five-Factor analysis.

4.1 Performance of Agri-food Portfolios

Our first set of results show the performance of the three agri-food portfolios against the market portfolio in the S&P 500 and STOXX 600 (Table 2). In the S&P 500, all industry segment portfolios underperform compared to the market portfolio. The composite portfolio of all agri-food stocks (*All_Agrifood_US*) is outperformed by the market by 77.06%. The Production and Processing segment (*Processing_US*) has underperformed particularly poorly, returning merely 51.23% compared to the market portfolio’s respective 189.75%. In the STOXX 600, the picture is qualitatively similar yet the magnitudes of underperformance are larger. For instance, the portfolio of food retailing and distribution (*Retail_EU*) companies underperforms by 89.6% compared to 33.62% by its US counterpart (*Retail_US*). Overall, these results indicate that agri-food portfolios in Europe tend to have a higher equity cost of capital than their US counterparts.

Comparing the Sharpe and Sortino ratios indicate that agri-food portfolios generally underperform in risk adjusted terms in both markets. At the $p < 0.5$ level, we reject the null hypothesis of equal Sharpe ratios using the JK and LW tests for most portfolios. Sortino ratios, which weight returns by their downside risk, also show that agri-food portfolios underperform. Underperformance of the agri-food sector differentiates it from other sectors with high asset stranding risk in future climate scenarios such as the fossil fuel sector which has typically shown equivalent performance or outperformance compared to the market portfolio (Plantinga & Scholtens, 2024).

We analyse the factors affecting the returns of each portfolio using the Fama–French Five Factor Model (Table 3). No portfolios show significant alphas, confirming the findings of the descriptive analysis—agri-food portfolios do not outperform the market. All portfolios show consistently significant factor loadings only on the market portfolio. The relatively low beta (< 1.0) of the majority of agri-food sector portfolios is relevant in considerations of divestment motivated by asset stranding risk. As defensive stocks, they may add considerable diversification benefits for investors, making divestment a much less desirable prospect than for stocks with betas closer to 1 such as fossil fuels (Trinks et al., 2018). This is particularly

Table 2. Performance of agri-food portfolios. Panel (a) reports the S&P 500 (USA); Panel (b) reports the STOXX 600 (Europe). Total and annualised returns are in percentages.

Panel (a): S&P 500 (USA)					
Portfolio Name	Total Return	Annualised Return	Annualised Risk	Sharpe Ratio	Sortino Ratio
Market_US	189.7537	21.495705	15.19666	0.3728848	0.6378532
Inputs_US	186.8294	21.335074	22.94381	0.2452858	0.4375183
Processing_US	51.2259	5.478232	13.43213	0.1148782	0.1762512
Distribution_US	156.1224	17.564203	14.97580	0.3140277	0.5953472
All_Agri-food_US	112.6851	12.413307	12.79185	0.2653542	0.4689157

Panel (b): STOXX 600 (Europe)					
Portfolio Name	Total Return	Annualised Return	Annualised Risk	Sharpe Ratio	Sortino Ratio
Market_EU	189.03600	21.610275	26.76245	0.2127705	0.5638097
Inputs_EU	162.70545	18.363447	30.30984	0.1617005	0.3287617
Processing_EU	57.01560	6.114397	14.86454	0.1155407	0.1741643
Retail_EU	99.48215	10.889168	15.18118	0.1973934	0.3549916
All_Agri-food_EU	74.53431	8.060814	13.09282	0.1714809	0.2703685

the case in climate scenarios which increase overall market volatility. Slight variations in betas between comparable portfolios in regions may pronounce this effect. For example, in the US, the inputs sector has a higher beta (0.99) compared to Europe (0.764). Divestment from this industry segment may therefore have a smaller impact on diversification costs in the US than in Europe.

4.2 Performance of Divested Portfolios

Next, we examine the risk-return profile of our portfolio divestment exercise in the S&P 500 and STOXX 600 (Table 4). This is an exercise analogous to divestment in the fossil fuel industry, which has been recently advanced by prominent activist groups such as Friends of the Earth International (2026). Looking strictly at returns, we find that divesting from different segments of the agri-food sector results in marginally higher returns and higher risk, with the effects more pronounced in the former. In absolute terms, the composite portfolio in both the US (*All_Agri-food_US_Div*) and Europe (*All_Agri-food_EU_Div*) returns 4.1% and 5.1% higher than the market respectively. In both markets, divestment from the Inputs segment underperforms very slightly.

However, the Sharpe and Sortino ratios indicate that these increases in return are compensated by equivalent increases in risk—divested portfolios perform no different in risk adjusted terms. These findings are consistent with those of Trinks et al. (2018) and Plantinga

Table 3. Fama–French Five Factor regressions for agri-food portfolios. Panel (a) reports US portfolios; Panel (b) reports EU portfolios. Standard errors in parentheses.

Panel (a): S&P 500 (USA)				
	Inputs_US	Processing_US	Retail_US	All_Agri-food_US
(Intercept)	0.004 (0.004)	-0.004 (0.003)	0.003 (0.003)	-0.000 (0.002)
mkt	0.985*** (0.100)	0.505*** (0.066)	0.706*** (0.069)	0.633*** (0.053)
smb	0.178 (0.173)	-0.147 (0.115)	-0.238* (0.119)	-0.172+ (0.091)
hml	0.360* (0.154)	0.077 (0.102)	-0.084 (0.106)	0.029 (0.081)
rmw	-0.006 (0.212)	0.501*** (0.141)	0.328* (0.146)	0.361** (0.112)
cma	0.378 (0.229)	0.324* (0.151)	0.240 (0.157)	0.293* (0.120)
Num. Obs.	115	115	115	115
R ²	0.580	0.464	0.530	0.623
R ² Adj.	0.561	0.439	0.509	0.606

Panel (b): STOXX 600 (Europe)				
	Inputs_EU	Processing_EU	Retail_EU	All_Agri-food_EU
(Intercept)	0.006 (0.008)	0.001 (0.003)	0.003 (0.003)	0.002 (0.002)
mkt	0.764*** (0.185)	0.750*** (0.066)	0.767*** (0.067)	0.749*** (0.048)
smb	0.623 (0.487)	-0.335+ (0.174)	0.031 (0.175)	-0.199 (0.126)
hml	0.338 (0.493)	-0.779*** (0.177)	-0.450* (0.178)	-0.672*** (0.128)
rmw	1.009 (0.748)	0.154 (0.268)	-0.253 (0.269)	0.044 (0.194)
cma	0.861 (0.870)	0.962** (0.311)	0.523+ (0.313)	0.847*** (0.225)
Num. Obs.	115	115	115	115
R ²	0.210	0.578	0.587	0.715
R ² Adj.	0.174	0.559	0.568	0.702

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 4. Performance of divested portfolios. Panel (a) reports the S&P 500 (USA); Panel (b) reports the STOXX 600 (Europe). Total and annualised returns are in percentages.

Panel (a): S&P 500 (USA)					
Portfolio Name	Total Return	Annualised Return	Annualised Risk	Sharpe Ratio	Sortino Ratio
Market_US_Div	189.75	21.49	15.19	0.37	0.63
Inputs_US_Div	189.68	21.48	15.19	0.37	0.63
Processing_US_Div	193.17	21.91	15.40	0.37	0.64
Retail_US_Div	190.42	21.57	15.32	0.37	0.63
All_Agri-food_US_Div	193.88	22.01	15.54	0.37	0.63
Panel (b): STOXX 600 (Europe)					
Portfolio Name	Total Return	Annualised Return	Annualised Risk	Sharpe Ratio	Sortino Ratio
Market_EU_Div	189.03	21.61	26.76	0.21	0.56
Inputs_EU_Div	189.02	21.60	26.79	0.21	0.56
Processing_EU_Div	192.95	22.09	27.51	0.21	0.56
Retail_EU_Div	190.17	21.75	26.95	0.21	0.56
All_Agri-food_EU_Div	194.16	22.25	27.76	0.21	0.56

and Scholtens (2021) in the energy sector who find no outperformance of portfolios divested from fossil fuels, but differ from Hunt and Weber (2019) who find that divested portfolios do indeed outperform. These results may, in part, be explained by the fact that in both markets the agri-food sector comprises a relatively small number of firms, and so, the divested portfolios are still highly diversified. Nonetheless, this result remains important from the perspective of asset stranding risk as it implies that the variation in the risk profile of the agri-food sector *against* the market is marginal.

Comparison of portfolios using the Fama–French Five Factor model (Table 5) indicates that all divested portfolios have a negligible positive alpha that is only significant in the USA. The risk factors show considerable variation in performance between US and European markets. In the US, divested portfolios closely track the market portfolio with statistically significant betas of near to, but less than 1 for all except the composite portfolio. All portfolios show significant factor loading on the Size (SMB) and Profitability factor (RMW). The latter lends support to the recent literature emphasising the role of real corporate profits in explaining stock returns (Asness et al., 2019; Liu et al., 2023; Novy-Marx, 2013; Salisu et al., 2025). However, in Europe, only the market factor is significant, with all divested portfolios showing statistically significant betas greater than 1. Divested portfolios are therefore generally riskier in Europe than in the USA.

Table 5. Fama–French Five Factor regressions for divested portfolios. Panel (a) reports US portfolios; Panel (b) reports EU portfolios. Standard errors in parentheses.

Panel (a): S&P 500 (USA)				
	Inputs_US_Div	Processing_US_Div	Retail_US_Div	All_Agri-food_Div
(Intercept)	0.003*** (0.000)	0.004*** (0.000)	0.003*** (0.000)	0.004*** (0.000)
mkt	0.983*** (0.010)	0.995*** (0.010)	0.989*** (0.010)	1.002*** (0.011)
smb	-0.123*** (0.018)	-0.119*** (0.018)	-0.119*** (0.018)	-0.117*** (0.018)
hml	0.005 (0.016)	0.005 (0.015)	0.008 (0.016)	0.005 (0.016)
rmw	0.061** (0.022)	0.051* (0.021)	0.055* (0.022)	0.046* (0.023)
cma	-0.001 (0.023)	-0.008 (0.023)	-0.005 (0.024)	-0.016 (0.024)
Num. Obs.	116	116	116	116
R ²	0.990	0.990	0.990	0.989
R ² Adj.	0.989	0.990	0.989	0.989
Panel (b): STOXX 600 (Europe)				
	Inputs_EU_Div	Processing_EU_Div	Retail_EU_Div	All_Agri-food_EU_Div
(Intercept)	0.008 (0.006)	0.009 (0.006)	0.008 (0.006)	0.009 (0.006)
mkt	1.154*** (0.135)	1.169*** (0.139)	1.159*** (0.136)	1.176*** (0.141)
smb	0.331 (0.355)	0.356 (0.367)	0.336 (0.358)	0.360 (0.372)
hml	-0.517 (0.359)	-0.500 (0.372)	-0.516 (0.362)	-0.501 (0.376)
rmw	-0.500 (0.545)	-0.515 (0.564)	-0.501 (0.549)	-0.521 (0.570)
cma	0.708 (0.634)	0.695 (0.656)	0.710 (0.639)	0.697 (0.663)
Num. Obs.	115	115	115	115
R ²	0.457	0.449	0.456	0.447
R ² Adj.	0.432	0.424	0.431	0.421

⁺ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

4.3 Best and Worst in Class Portfolios

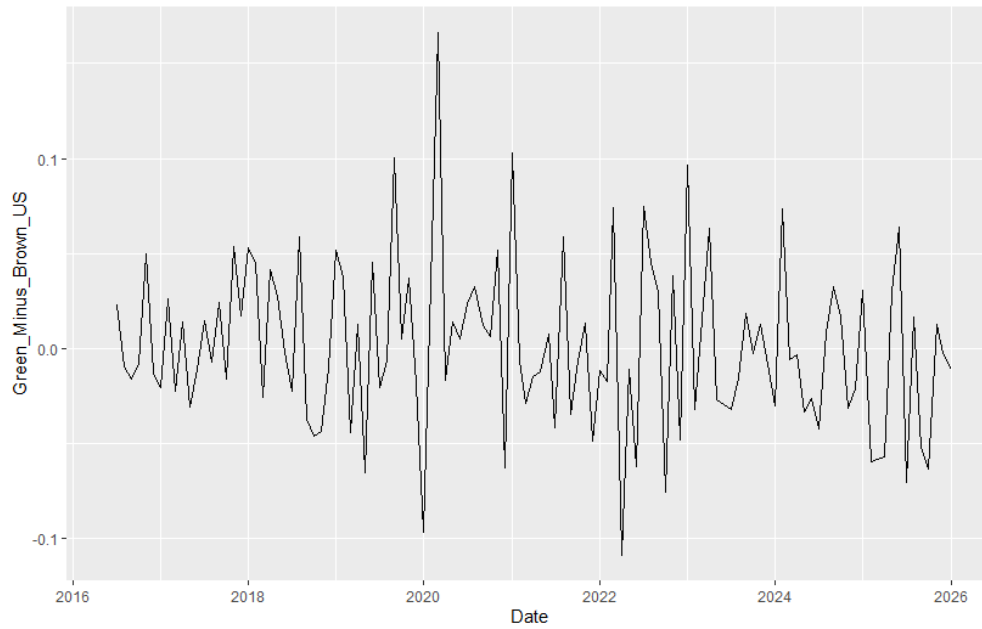
Finally, we analyse the return spread between our best and worst in class portfolio, which we term the Green Minus Brown (GMB) spread, in the spirit of Pástor et al. (2022). Should investors penalise brown firms (marked as the bottom quartile of firms by Scope 3 emissions), or conversely reward green firms (marked as the top quartile of firms by Scope 3 emissions), within the agri-food sector we would expect this spread to be positive. Figures 2 and 2 plot the GMB spread in the US and EU respectively. The trend line is highly volatile but oscillates around a near 0 mean over the entire sample. The standard deviation is 0.044 in both markets.

The descriptive statistics show that the mean spread is positive but extremely small over our sample period in both markets (Table 6). In the US the median is negative, whereas in the EU it sits higher than the mean. However, the small magnitudes of these values imply that they cannot be interpreted as particularly significant for an investor wishing to capitalise on a return spread over a long time period. Volatility measured by the standard deviation is equal in both markets. The results suggest overall that there is no persistent outperformance of higher emitting firms, indicating that investors are not rewarding companies with lower asset stranding risk in rapid decarbonisation scenarios within the agri-food sector.

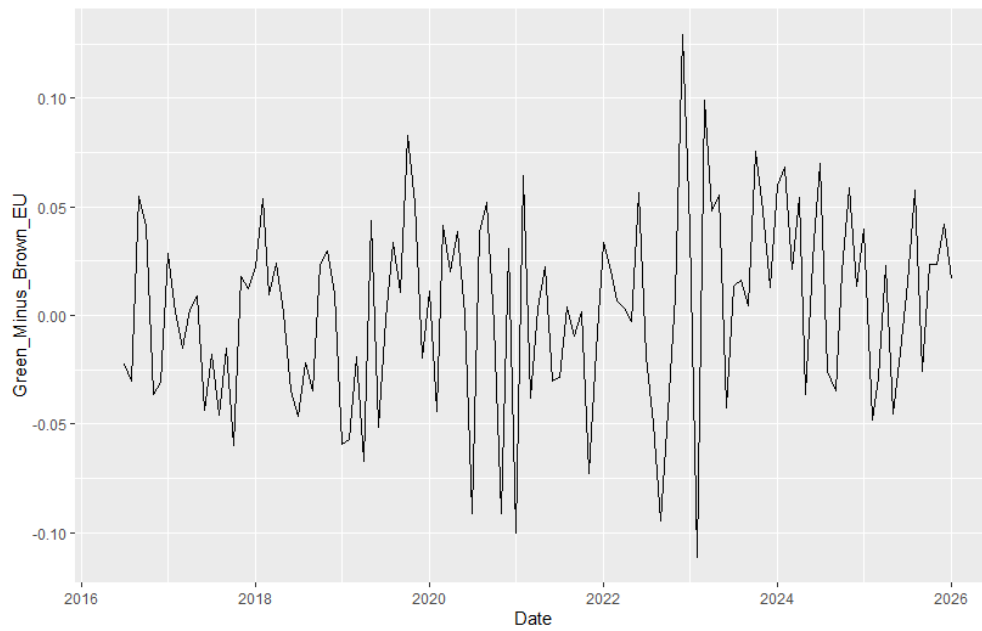
Table 6. Green Minus Brown (GMB) Descriptive Statistics. All values report the excess return in percentages of the green portfolio over the brown. Subscript US denotes the S&P 500 and EU denotes the STOXX 600.

	GMB_US (Total)	GMB_EU (Total)	GMB_US (2016–2021)	GMB_EU (2016–2021)	GMB_US (2021–2026)	GMB_EU (2021–2026)
Mean	0.001	0.002	0.009	0.001	−0.007	0.009
Median	−0.006	0.008	0.007	−0.005	−0.011	0.013
Standard Deviation	0.044	0.044	0.044	0.041	0.043	0.046
Skew	0.08	0.27	−0.10	0.39	0.19	0.17
Kurt.	3.33	3.26	2.21	3.18	3.90	3.31
Max	0.166	0.129	0.166	0.083	0.097	0.129
Min	−0.108	−0.111	−0.097	0.01	−0.108	−0.111

We further attempt to tease out time varying properties of this spread, by splitting the sample into two time periods, 2016–2020 and 2021–2026. We choose to separate our sample in this way as it is plausible that, as scientific understanding of the risks associated with the ecological crisis has increased in the past decade, the latter time period may see investors starting to price them. In the latter period, we note a slight deviation in the trend between the two markets, with the spread increasing in Europe, and falling to negative in the US. While this magnitude is small, this difference may be interpreted as a result of increased



(a)



(b)

Figure 2. Green-Minus-Brown Spread for the US (a) and EU (b) markets. Green-Minus-Brown is the percentage monthly return on a portfolio long on the top quartile of firms by Scope 3 emissions intensity and short on the bottom quartile beginning in June 2016. The Y axis measures the percentage return difference.

environmental compliance measures in the EU as compared to the US, i.e. investors are incentivised to reward greener stocks more in the former than the latter.

Finally, we test for the significance of the GMB factor by adding it as an additional control in the Fama–French Five Factor asset pricing model. We test its significance in the two portfolios which divest from the entire agri-food sector (All_Agri-food_Div_US and All_Agri-food_Div_EU). The results (Table 7) show that the GMB factor is not significant in either market, indicating that the Scope 3 emissions intensity of a portfolio does not appear as a relevant risk factor for investors in the agri-food sector.

Table 7. Fama–French Five Factor regressions with the Green-Minus-Brown (GMB) factor. Models 1 and 3 are US portfolios; Models 2 and 4 are EU portfolios. Models 3 and 4 add the GMB factor (gmb). Standard errors in parentheses.

	All_Agri-Food_US_Div	All_Agri-Food_EU_Div	All_Agri-Food_US_Div	All_AgriFood_EU_Div
(Intercept)	0.004*** (0.000)	0.009 (0.006)	0.004*** (0.000)	0.009 (0.006)
mkt	1.002*** (0.011)	1.176*** (0.141)	1.002*** (0.011)	1.179*** (0.142)
smb	-0.117*** (0.018)	0.360 (0.372)	-0.116*** (0.019)	0.354 (0.373)
hml	0.005 (0.016)	-0.501 (0.376)	0.004 (0.017)	-0.505 (0.378)
rmw	0.046* (0.023)	-0.521 (0.570)	0.048* (0.023)	-0.554 (0.576)
cma	-0.016 (0.024)	0.697 (0.663)	-0.015 (0.025)	0.667 (0.669)
gmb			0.002 (0.007)	0.040 (0.081)
Num. Obs.	116	115	115	115
R ²	0.989	0.447	0.990	0.448
R ² Adj.	0.989	0.421	0.989	0.417

⁺ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

5 Discussion and Conclusion

In plausible future climate scenarios, the agri-food sector bears considerable physical and transition risks which threaten investors with potentially devalued financial assets. Our study advances the case that it is important to investigate the materiality of these risks, as it is a key determinant of investment in sustainable food systems (Walton et al., 2025). We explored this idea through the case study of divestment, echoing recent activist calls for greater

1
2
3
4 attention to sustainable corporate governance in the agri-food sector. Our findings suggest
5 that divestment from the agri-food sector has had little impact on portfolio performance in
6 the decade since the Paris Agreement in the USA or in Europe.
7

8
9 Where the wider literature typically finds equivalent performance or outperformance of
10 stocks at risk of asset stranding such as fossil fuels (Plantinga & Scholtens, 2021; Trinks
11 et al., 2018), we find that agri-food sector underperformed against the market in risk adjusted
12 terms with statistically significant underperformance observable in the food production and
13 processing segment. We find that the return spread on a best-in-class portfolio against a
14 worst-in-class portfolio is small but generally positive. The gap between the return spread
15 in the US and Europe widens in recent years. The spread between high and low emissions
16 intensity firms is not significant in explaining divested portfolio returns. Our results indicate
17 that asset stranding risks do not appear to be materially affecting portfolio returns in the
18 agri-food sector. However, the sector shows some differences with other sectors at risk of
19 asset stranding such as fossil fuels.
20

21
22 Theoretically, these results can be plausibly interpreted through the lens of efficient
23 markets, suggesting that investors are aware of and already pricing in the asset stranding
24 risks associated with the food system against the market. However, we suggest that these
25 results are better interpreted through the more general evolutionary lens of adaptive markets.
26 While it is no doubt the case that there is insufficient information on the asset stranding
27 risks particular to the agri-food system, it is not clear that, were better information available,
28 financial institutions would be well equipped to price them. Indeed, the opposite appears
29 to be the want of the finance industry—to *reduce* the complexity involved in environmental
30 disclosure into fewer, more streamlined, metrics. Consequently, there is sufficient reason to
31 doubt that increasing environmental disclosure will *ipso facto* allow investors to reflect equity
32 devaluations due to climate scenarios in current prices (Christophers, 2017; Silver, 2018). In
33 an adaptive framework, if financial institutions are to divest at scale, there must be far more
34 than simply environmental disclosures in place for them to be able to do so.
35

36
37 On the one hand, our findings may be good news for social movements and institutional
38 investors aiming to divest from the agri-food system due to perceived risk of asset stranding.
39 Had they divested from the industry in the recent past, their risk-adjusted return would
40 likely be unaffected. Indeed, given the small number of agri-food stocks on both the S&P 500
41 and STOXX 600, there is little to be lost from a divestment exercise. However, there are
42 many reasons to be skeptical that such a project will materialise at a scale analogous to the
43 fossil fuel divestment movement, many of which can provide the basis for future research.
44 Some of the most prominent follow.
45

46
47 First, and most importantly, there are no generally agreed on frameworks to assess “green”
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65

1
2
3
4 and “brown” stocks in the agri-food sector. In contrast to the energy sector, where a typical
5 framing can credibly position “fossil free” and “fossil fuel” as a realistic portfolio choice for a
6 concerned investor, there are few frameworks to assess the same in the food system despite
7 the growing scientific awareness of its impacts. Relatedly, while divestment movements in the
8 energy sector typically argue for re-investment in green industry such as in renewable energy
9 companies (DivestInvest, 2021; Hunt & Weber, 2019), our survey of the agri-food sector
10 found few analogous green indices into which an investor could re-allocate funds. Moreover,
11 these indices are not generally issued by the large commercial index providers. The credibility
12 of a divestment campaign in the agri-food sector may turn on this point as institutional
13 investors are likely to question how divested funds should be re-invested. Taking an adaptive
14 markets view, this is fundamentally a policy question for governments to identify and support
15 the growth of fledgling industries in sustainable food production. While there is growing
16 recognition of the role of the public sector in supporting transitions in the energy sector
17 (Christophers, 2025; Grubb et al., 2017), this remains an underdeveloped theme in the food
18 literature.
19

20
21 Second, there remains a problem of firm level data on environmental risk most relevant
22 for the food sector which remain sparse, for financial institutions as well as regulators.
23 Only recently are they becoming components of environmental reporting standards through
24 voluntary mechanisms such as the Taskforce for Nature Related Financial Disclosures (TFND)
25 and legally through the Corporate Sustainability Reporting Directive (CSRD) in the EU.
26 The lack of good quality data with broad coverage limits both researchers and practitioners
27 from clearly developing frameworks to screen and engage with the most environmentally
28 harmful agri-food corporations, potentially leading to overly simplistic governance strategies.
29 Moreover, the lack of data limits assessing the environmental risks specific to the food system
30 and pricing these in. As noted above, this has some role to play in explaining the lack of
31 impact of divestment on risk-adjusted returns we find in our analysis—investors may simply
32 not have the tools to credibly do so. Future research can use novel identification strategies
33 such as the text or news based approaches used by Engle et al. (2020) and Giglio et al.
34 (2023) as proxies for firm level environmental risk, as well as forward looking exposure based
35 modelling approaches such as Battiston et al. (2017, 2021).
36

37
38 Third, divestment from the agri-food sector due to asset stranding risk would require
39 institutional investors to take a more active role in portfolio construction than indexing against
40 a benchmark. Given divestment has little direct impact, a passive, index oriented investor
41 would likely see little reason to affect the diversification of their portfolios, particularly given
42 the low beta of agri-food stocks. Moreover, divestment from the agri-food sector is unlikely to
43 occur in isolation. Environmentally conscious investors are more likely to exclude agri-food
44
45
46
47
48
49
50
51
52
53
54
55
56
57
58
59
60
61
62
63
64
65

as a component of a wider risk management strategy, likely including other perceived risky stocks such as coal or oil and gas. Further research should explore how different divestment strategies may work in concert or contradiction with each other. More structurally, in an institutional environment in which passive index investing is the norm, decision rules which encourage such divestment from large institutional investors are likely to remain marginal as they do not facilitate the pricing of asset stranding risk. We suggest that agent based modelling approaches, which incorporate the interactions of heterogeneous decision rules, is a useful methodological framework to advance research on investment practices more aligned with sustainable transitions (Axtell & Farmer, 2025; Battiston et al., 2016).

Finally, and more generally, our study underscores the need to understand asset stranding risk as a primary causal variable in explaining the economics of large scale industrial transitions to sustainability generally, and in the food sector specifically (Caldecott, 2017; Semieniuk et al., 2022; Walton et al., 2025). Put simply, the (lack of) materiality of such risks drastically affect the scale and tempo of investment and divestment in sustainable production. Industrial transitions are generally associated with considerable financial market volatility through the restructuring of industrial organisation—growing pains of any process of “creative destruction” (Mazzucato & Perez, 2023; Perez, 2002). That we do not observe these dynamics in our sample of agri-food corporations should be cause for concern for financial supervisors and regulators concerned with financial stability, as asset stranding risk may come to bear with far more significant impact for financial stability in the future if not dealt with proactively (Battiston et al., 2017; Caldecott et al., 2021; Carney, 2015). Ultimately, a framework for green finance in the food system which encourages divestment from assets exposed to environmental risk by large financial institutions can only materialise through proactive “market-shaping” by financial supervisors (Ryan-Collins, 2019).

References

- Ackerman, F. (2018). *Worst-Case Economics: Extreme Events in Climate and Finance*. Anthem Press.
- Alessi, L., Ossola, E., & Panzica, R. (2021). What greenium matters in the stock market? The role of greenhouse gas emissions and environmental disclosures. *Journal of Financial Stability*, 54, 100869. <https://doi.org/10.1016/j.jfs.2021.100869>
- Ameli, N., Drummond, P., Bisaro, A., Grubb, M., & Chenet, H. (2020). Climate finance and disclosure for institutional investors: Why transparency is not enough. *Climatic Change*, 160(4), 565–589. <https://doi.org/10.1007/s10584-019-02542-2>

- Angus, I. (2016, July). *Facing the Anthropocene: Fossil Capitalism and the Crisis of the Earth System* [Google-Books-ID: R8XOCwAAQBAJ]. NYU Press.
- Ansar, A., Caldecott, B., & Tilbury, J. (2013). Stranded assets and the fossil fuel divestment campaign: What does divestment mean for the valuation of fossil fuel assets? [Stub entry: please verify full bibliographic details.]. *Smith School of Enterprise and the Environment, University of Oxford*.
- Ardia, D., Bluteau, K., Boudt, K., & Inghelbrecht, K. (2023). Climate Change Concerns and the Performance of Green vs. Brown Stocks. *Management Science*, 69(12), 7607–7632. <https://doi.org/10.1287/mnsc.2022.4636>
- Asness, C. S., Frazzini, A., & Pedersen, L. H. (2019). Quality minus junk. *Review of Accounting Studies*, 24(1), 34–112. <https://doi.org/10.1007/s11142-018-9470-2>
- Axtell, R. L., & Farmer, J. D. (2025). Agent-Based Modeling in Economics and Finance: Past, Present, and Future. *Journal of Economic Literature*, 63(1), 197–287. <https://doi.org/10.1257/jel.20221319>
- Ayling, J., & Gunningham, N. (2017). Non-state governance and climate policy: The fossil fuel divestment movement. *Climate Policy*, 17(2), 131–149. <https://doi.org/10.1080/14693062.2015.1094729>
- Battiston, S., Dafermos, Y., & Monasterolo, I. (2021). Climate risks and financial stability. *Journal of Financial Stability*, 54, 100867. <https://doi.org/10.1016/j.jfs.2021.100867>
- Battiston, S., Farmer, J. D., Flache, A., Garlaschelli, D., Haldane, A. G., Heesterbeek, H., Hommes, C., Jaeger, C., May, R., & Scheffer, M. (2016). Complexity theory and financial regulation. *Science*, 351(6275), 818–819. <https://doi.org/10.1126/science.aad0299>
- Battiston, S., Mandel, A., Monasterolo, I., Schütze, F., & Visentin, G. (2017). A climate stress-test of the financial system. *Nature Climate Change*, 7(4), 283–288. <https://doi.org/10.1038/nclimate3255>
- Bolton, P., & Kacperczyk, M. (2021). Do investors care about carbon risk? *Journal of Financial Economics*, 142(2), 517–549. <https://doi.org/10.1016/j.jfineco.2021.05.008>
- Bolton, P., & Kacperczyk, M. (2023). Global Pricing of Carbon-Transition Risk. *The Journal of Finance*, 78(6), 3677–3754. <https://doi.org/10.1111/jofi.13272>
- Braungardt, S., van den Bergh, J., & Dunlop, T. (2019). Fossil fuel divestment and climate change: Reviewing contested arguments. *Energy Research & Social Science*, 50, 191–200. <https://doi.org/10.1016/j.erss.2018.12.004>
- Bray, D., & Poston, T. M. (2024). The Methane Majors: Climate Change and Animal Agriculture in U.S. Courts. *Columbia Journal of Environmental Law*, 49(Symposium Issue), 145–248. <https://doi.org/10.52214/cjel.v49iS.12548>

- Caldecott, B. (2017). Introduction to special issue: Stranded assets and the environment. *Journal of Sustainable Finance & Investment*, 7(1), 1–13. <https://doi.org/10.1080/20430795.2016.1266748>
- Caldecott, B. (Ed.). (2018, May). *Stranded Assets and the Environment: Risk, Resilience and Opportunity*. Routledge. <https://doi.org/10.4324/9781315651606>
- Caldecott, B., Clark, A., Koskelo, K., Mulholland, E., & Hickey, C. (2021). Stranded Assets: Environmental Drivers, Societal Challenges, and Supervisory Responses. *Annual Review of Environment and Resources*, 46(Volume 46, 2021), 417–447. <https://doi.org/10.1146/annurev-environ-012220-101430>
- Campiglio, E., Daumas, L., Monnin, P., & Von Jagow, A. (2023). Climate-related risks in financial assets. *Journal of Economic Surveys*, 37(3), 950–992. <https://doi.org/10.1111/joes.12525>
- Carney, M. (2015). *Breaking the Tragedy of the Horizon – climate change and financial stability* (tech. rep.). Bank of England. Retrieved May 8, 2026, from <https://www.bankofengland.co.uk/-/media/boe/files/speech/2015/breaking-the-tragedy-of-the-horizon-climate-change-and-financial-stability.pdf>
- Christophers, B. (2017). Climate Change and Financial Instability: Risk Disclosure and the Problematics of Neoliberal Governance. *Annals of the American Association of Geographers*, 107(5), 1108–1127. <https://doi.org/10.1080/24694452.2017.1293502>
- Christophers, B. (2025, June). *The Price is Wrong: Why Capitalism Won't Save the Planet* [Google-Books-ID: HKUkEQAAQBAJ]. Verso Books.
- Cochrane, J. H. (1994). *Asset pricing* [Stub entry: the manuscript cites Cochrane (1994); the .bib contains Cochrane (2009). Please verify intended edition.]. Princeton University Press.
- Crippa, M., Solazzo, E., Guizzardi, D., Monforti-Ferrario, F., Tubiello, F. N., & Leip, A. (2021). Food systems are responsible for a third of global anthropogenic GHG emissions. *Nature Food*, 2(3), 198–209. <https://doi.org/10.1038/s43016-021-00225-9>
- Dasgupta, P. (2021, August). *The Economics of Biodiversity: The Dasgupta Review* (tech. rep.). HM Treasury. Retrieved May 19, 2026, from <https://www.gov.uk/government/publications/final-report-the-economics-of-biodiversity-the-dasgupta-review>
- DivestInvest. (2021). *Invest-Divest: A Decade of Progress towards a Just Climate Future* (tech. rep.). Retrieved March 3, 2026, from https://www.divestinvest.org/wp-content/uploads/2021/10/Divest-Invest-Program-FINAL10-26_B.pdf
- Dordi, T., Krueger, R., Duff, P., & Brodsky, G. (2025). Framing divestment: A decade of media discourse on fossil fuel divestment in higher education. *Climatic Change*, 178(6), 121. <https://doi.org/10.1007/s10584-025-03962-z>

- 1
- 2
- 3
- 4 Edmans, A. (2024). Rational sustainability. *Journal of Applied Corporate Finance*, 36(2),
- 5 8–15. <https://doi.org/10.1111/jacf.12609>
- 6
- 7 Engle, R. F., Giglio, S., Kelly, B., Lee, H., & Stroebe, J. (2020). Hedging Climate Change News.
- 8 *The Review of Financial Studies*, 33(3), 1184–1216. <https://doi.org/10.1093/rfs/hhz072>
- 9
- 10 European Union. (2025, December). Regulation (EU) 2023/1115 of the European Parliament
- 11 and of the Council of 31 May 2023 on the making available on the Union market
- 12 and the export from the Union of certain commodities and products associated with
- 13 deforestation and forest degradation and repealing Regulation (EU) No 995/2010
- 14 [Legislative Body: OP_DATPRO]. Retrieved March 6, 2026, from [http://data.europa.](http://data.europa.eu/eli/reg/2023/1115/2025-12-26)
- 15 [eu/eli/reg/2023/1115/2025-12-26](http://data.europa.eu/eli/reg/2023/1115/2025-12-26)
- 16
- 17 Ewert, F., Baatz, R., & Finger, R. (2023). Agroecology for a Sustainable Agriculture and Food
- 18 System: From Local Solutions to Large-Scale Adoption. *Annual Review of Resource*
- 19 *Economics*, 15(Volume 15, 2023), 351–381. [https://doi.org/10.1146/annurev-resource-](https://doi.org/10.1146/annurev-resource-102422-090105)
- 20 [102422-090105](https://doi.org/10.1146/annurev-resource-102422-090105)
- 21
- 22 Falkner, R. (2016). The paris agreement and the new logic of international climate politics
- 23 [Stub entry: please verify full bibliographic details.]. *International Affairs*, 92(5),
- 24 1107–1125.
- 25
- 26 Fama, E. F. (1970). Efficient capital markets: A review of theory and empirical work [Stub
- 27 entry: please verify full bibliographic details.]. *The Journal of Finance*, 25(2), 383–417.
- 28
- 29 Fama, E. F., & French, K. R. (2016). Dissecting Anomalies with a Five-Factor Model. *The*
- 30 *Review of Financial Studies*, 29(1), 69–103. <https://doi.org/10.1093/rfs/hhv043>
- 31
- 32 FAO. (2023, December). *Achieving SDG 2 without breaching the 1.5 °C threshold: A global*
- 33 *roadmap, Part 1* (tech. rep.). FAO. <https://doi.org/10.4060/cc9113en>
- 34
- 35 Farmer, J. D., & Lo, A. W. (1999). Frontiers of finance: Evolution and efficient markets.
- 36 *Proceedings of the National Academy of Sciences*, 96(18), 9991–9992. [https://doi.org/](https://doi.org/10.1073/pnas.96.18.9991)
- 37 [10.1073/pnas.96.18.9991](https://doi.org/10.1073/pnas.96.18.9991)
- 38
- 39 Forest500. (2024). *Forest 500 Annual Report 2024* (tech. rep.). Forest 500. Retrieved March
- 40 6, 2026, from [https://forest500.org/wp-content/uploads/2024/02/Forest500_Annual-](https://forest500.org/wp-content/uploads/2024/02/Forest500_Annual-Report-2024_Final.pdf)
- 41 [Report-2024_Final.pdf](https://forest500.org/wp-content/uploads/2024/02/Forest500_Annual-Report-2024_Final.pdf)
- 42
- 43 Friends of the Earth International. (2026). *Introducing the Group of 100* (tech. rep.). Retrieved
- 44 April 27, 2026, from [https://foe.org/wp-content/uploads/2026/04/Introducing-the-](https://foe.org/wp-content/uploads/2026/04/Introducing-the-Group-of-100.pdf)
- 45 [Group-of-100.pdf](https://foe.org/wp-content/uploads/2026/04/Introducing-the-Group-of-100.pdf)
- 46
- 47 FSDA. (2024). *Finance Sector Deforestation Action (FSDA) Progress Report* (tech. rep.).
- 48 Retrieved March 6, 2026, from [https://www.climatechampions.net/media/cnjfueq/](https://www.climatechampions.net/media/cnjfueq/fsda-progress-report-june-2024.pdf)
- 49 [fsda-progress-report-june-2024.pdf](https://www.climatechampions.net/media/cnjfueq/fsda-progress-report-june-2024.pdf)
- 50
- 51
- 52
- 53
- 54
- 55
- 56
- 57
- 58
- 59
- 60
- 61
- 62
- 63
- 64
- 65

- Gaupp, F., Hall, J., Hochrainer-Stigler, S., & Dadson, S. (2019). Changing risks of simultaneous global breadbasket failure [Stub entry: please verify full bibliographic details.]. *Nature Climate Change*.
- Giglio, S., Kelly, B., & Stroebel, J. (2021). Climate finance [Stub entry: the manuscript cites Giglio et al. (2021); the .bib contains Giglio et al. (2023, Biodiversity Risk). Please verify.]. *Annual Review of Financial Economics*, 13, 15–36.
- Giglio, S., Kuchler, T., Stroebel, J., & Zeng, X. (2023, April). Biodiversity Risk. <https://doi.org/10.3386/w31137>
- Gilbert, E., & Strugnell, D. (2010). Does survivorship bias really matter? An empirical investigation into its effects on the mean reversion of share returns on the JSE (1984–2007). *Investment Analysts Journal*, 39(72), 31–42. <https://doi.org/10.1080/10293523.2010.11082521>
- Gliessman, S. R. (2021, February). *Package Price Agroecology: The Ecology of Sustainable Food Systems, Third Edition* [Google-Books-ID: rYUfEAAAQBAJ]. CRC Press.
- Global Witness. (2025). *Banks make \$26 billion in a decade of financing deforesting companies* (tech. rep.). Retrieved March 6, 2026, from https://gw.hacdn.io/media/documents/Banks_make_26_billion_in_a_decade_of_financing_deforesting_companies_Oct_2025.pdf
- GlobalCanopy.org. (2026). Deforestation-Free Finance: Finance Sector Roadmap. Retrieved March 6, 2026, from <https://guidance.globalcanopy.org/roadmap/>
- Gourdji, S., Pandey, A., & Peiris, N. (2025). *White Paper: Impacts of climate change on globalwheat production and supply chains* (tech. rep.). Moody's. Retrieved May 20, 2026, from https://www.moody's.com/web/en/us/insights/resources/2024_AgRisk_Climate_Change_Wheat_Prod.pdf
- Greenbank. (2024). *Sustainable food systems and systemic risks: What role does the Investor Coalition on Food Policy play to promote health and resilience?* (Tech. rep.). Retrieved March 6, 2026, from https://www.greenbankinvestments.com/sites/greenbankinvestments.com/files/literature/pdfs/2024-11/greenbank_sustainable_food_systems_and_systemic_risks_investor_coalition_on_food_policy_final.pdf
- Grubb, M., McDowall, W., & Drummond, P. (2017). On order and complexity in innovations systems: Conceptual frameworks for policy mixes in sustainability transitions. *Energy Research & Social Science*, 33, 21–34. <https://doi.org/10.1016/j.erss.2017.09.016>
- Hall, S., Foxon, T. J., & Bolton, R. (2017). Investing in low-carbon transitions: Energy finance as an adaptive market. *Climate Policy*, 17(3), 280–298. <https://doi.org/10.1080/14693062.2015.1094731>

- Harwatt, H., Ripple, W. J., Chaudhary, A., Betts, M. G., & Hayek, M. N. (2020). Scientists call for renewed Paris pledges to transform agriculture. *The Lancet Planetary Health*, 4(1), e9–e10. [https://doi.org/10.1016/S2542-5196\(19\)30245-1](https://doi.org/10.1016/S2542-5196(19)30245-1)
- Hoffmann, V. H., & Busch, T. (2008). Corporate Carbon Performance Indicators. *Journal of Industrial Ecology*, 12(4), 505–520. <https://doi.org/10.1111/j.1530-9290.2008.00066.x>
- Hong, H., Li, F. W., & Xu, J. (2019). Climate risks and market efficiency. *Journal of Econometrics*, 208(1), 265–281. <https://doi.org/10.1016/j.jeconom.2018.09.015>
- Hong, H., Lim, T., & Stein, J. C. (2000). Bad News Travels Slowly: Size, Analyst Coverage, and the Profitability of Momentum Strategies. *The Journal of Finance*, 55(1), 265–295. <https://doi.org/10.1111/0022-1082.00206>
- Huang, X., & Zhai, P. (2021). The paris agreement and global attention to climate change [Stub entry: please verify full bibliographic details.].
- Hultgren, A., Carleton, T., Delgado, M., Gergel, D. R., Greenstone, M., Houser, T., Hsiang, S., Jina, A., Kopp, R. E., Malevich, S. B., McCusker, K. E., Mayer, T., Nath, I., Rising, J., Rode, A., & Yuan, J. (2025). Impacts of climate change on global agriculture accounting for adaptation. *Nature*, 642(8068), 644–652. <https://doi.org/10.1038/s41586-025-09085-w>
- Hunt, C., & Weber, O. (2019). Fossil Fuel Divestment Strategies: Financial and Carbon-Related Consequences. *Organization & Environment*, 32(1), 41–61. <https://doi.org/10.1177/1086026618773985>
- Hunt, C., Weber, O., & Dordi, T. (2018). A comparative analysis of the anti-Apartheid and fossil fuel divestment campaigns. In *Stranded Assets*. Routledge.
- Intergovernmental Panel on Climate Change. (n.d.). *AR6 Synthesis Report Climate Change 2023* (tech. rep.). Retrieved March 2, 2026, from <https://www.ipcc.ch/assessment-report/ar6/>
- IPBES. (2019, November). *Summary for policymakers of the global assessment report on biodiversity and ecosystem services* (tech. rep.). Zenodo. <https://doi.org/10.5281/zenodo.3553579>
- Jobson, J. D., & Korkie, B. M. (1981). Performance hypothesis testing with the sharpe and treynor measures [Stub entry: please verify full bibliographic details.]. *The Journal of Finance*, 36(4), 889–908.
- Jones, A., Bedenham, G., Goldman, M., Ranchin, A., Spencer, N., & Trim, I. (2026). *Planetary Solvency: Tipping into the wild unknown* (tech. rep.). Institute and Faculty of Actuaries. Retrieved April 30, 2026, from <https://actuaries.org.uk/news-and-media-releases/news-articles/2026/apr/30-apr-26-planetary-solvency-tipping-into-the-wild-unknown/>

- Kesse-Guyot, E., Allès, B., Brunin, J., Fouillet, H., Dussiot, A., Berthy, F., Perraud, E., Hercberg, S., Julia, C., Mariotti, F., Deschasaux-Tanguy, M., Srouf, B., Lairon, D., Pointereau, P., Baudry, J., & Touvier, M. (2023). Environmental impacts along the value chain from the consumption of ultra-processed foods. *Nature Sustainability*, 6(2), 192–202. <https://doi.org/10.1038/s41893-022-01013-4>
- Lambin, E. F., & Furumo, P. R. (2023). Deforestation-Free Commodity Supply Chains: Myth or Reality? *Annual Review of Environment and Resources*, 48(Volume 48, 2023), 237–261. <https://doi.org/10.1146/annurev-environ-112321-121436>
- Ledoit, O., & Wolf, M. (2008). Robust performance hypothesis testing with the sharpe ratio [Stub entry: please verify full bibliographic details.]. *Journal of Empirical Finance*, 15(5), 850–859.
- Lesk, C., Anderson, W., Rigden, A., Coast, O., Jägermeyr, J., McDermid, S., Davis, K. F., & Konar, M. (2022). Compound heat and moisture extreme impacts on global crop yields under climate change. *Nature Reviews Earth & Environment*, 3(12), 872–889. <https://doi.org/10.1038/s43017-022-00368-8>
- Liu, Y., Yang, B., & Su, Y. (2023). Changes in firm profitability, heterogeneous investor beliefs, and stock returns. *Journal of Management Science and Engineering*, 8(2), 258–272. <https://doi.org/10.1016/j.jmse.2022.10.004>
- Lo, A. W. (2017, April). *Adaptive Markets: Financial Evolution at the Speed of Thought*. Princeton University Press. Retrieved May 27, 2026, from <https://www.degruyterbrill.com/document/doi/10.1515/9781400887767-007/html?lang=en>
- Lo, A. W., & Zhang, R. (2024, January). *The Adaptive Markets Hypothesis: An Evolutionary Approach to Understanding Financial System Dynamics* [Google-Books-ID: PEnzEAAAQBAJ]. Oxford University Press.
- Lobe, S., & Walkshäusl, C. (2016). Vice versus virtue investing around the world. *Review of Managerial Science*, 10(2), 303–344. <https://doi.org/10.1007/s11846-014-0147-3>
- Mandouri, J., Onat, N. C., Kucukvar, M., Jabbar, R., Al-Quradaghi, S., Al-Thani, S., & Kazançoğlu, Y. (2025). Carbon footprint of food production: A systematic review and meta-analysis. *Scientific Reports*, 15(1), 35630. <https://doi.org/10.1038/s41598-025-19476-8>
- Markowitz, H. (1952). Portfolio Selection. *The Journal of Finance*, 7(1), 77–91. <https://doi.org/10.2307/2975974>
- Mazzucato, M., & Perez, C. (2023). Innovation as growth policy [Stub entry: please verify full bibliographic details.].
- Mehrabi, Z. (2026). Bearing the costs of a climate-safe food transition. *Nature Food*, 7(1), 11–12. <https://doi.org/10.1038/s43016-025-01288-8>

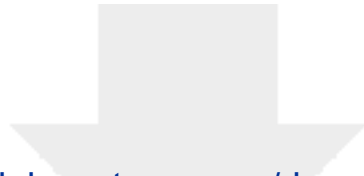
- Midgeley, A., Smith-Anthony, A., Abreu, M., Folland, I., Foldes, A., Baer, M., Rollo, A., & Akoka, A. (2021). *RISKY FINANCE: The UK financial sector's role in financing deforestation and conversion* (tech. rep.). https://www.wwf.org.uk/sites/default/files/2021-09/WWF_2021_Risky%20Finance%20Report.pdf
- Monasterolo, I. (2020). Climate Change and the Financial System. *Annual Review of Resource Economics*, 12(Volume 12, 2020), 299–320. <https://doi.org/10.1146/annurev-resource-110119-031134>
- Monasterolo, I., & de Angelis, L. (2020). Blind to carbon risk? An analysis of stock market reaction to the Paris Agreement. *Ecological Economics*, 170, 106571. <https://doi.org/10.1016/j.ecolecon.2019.106571>
- Norges Bank Investment Management. (2021, December). Decisions on observation and exclusion. Retrieved March 6, 2026, from <https://www.nbim.no/en/news-and-insights/the-press/press-releases/2021/decisions-on-observation-and-exclusion2/>
- Novy-Marx, R. (2013). The other side of value: The gross profitability premium. *Journal of Financial Economics*, 108(1), 1–28. <https://doi.org/10.1016/j.jfineco.2013.01.003>
- Pástor, L., Stambaugh, R. F., & Taylor, L. A. (2021). Sustainable investing in equilibrium. *Journal of Financial Economics*, 142(2), 550–571. <https://doi.org/10.1016/j.jfineco.2020.12.011>
- Pástor, L., Stambaugh, R. F., & Taylor, L. A. (2022). Dissecting green returns. *Journal of Financial Economics*, 146(2), 403–424. <https://doi.org/10.1016/j.jfineco.2022.07.007>
- Perez, C. (2002). *Technological revolutions and financial capital: The dynamics of bubbles and golden ages* [Stub entry: please verify full bibliographic details.]. Edward Elgar Publishing.
- Plantinga, A., & Scholtens, B. (2021). The financial impact of fossil fuel divestment. *Climate Policy*, 21(1), 107–119. <https://doi.org/10.1080/14693062.2020.1806020>
- Plantinga, A., & Scholtens, B. (2024). The finance perspective on fossil fuel divestment. *Current Opinion in Environmental Sustainability*, 66, 101394. <https://doi.org/10.1016/j.cosust.2023.101394>
- Portfolio Earth. (2021). *Bankrolling Pesticides* (tech. rep.). Retrieved March 6, 2026, from https://portfolio.earth/wp-content/uploads/2021/12/Portfolio-Earth_Bankrolling-Pesticides.pdf
- Rockström, J., Edenhofer, O., Gaertner, J., & DeClerck, F. (2020). Planet-proofing the global food system. *Nature Food*, 1(1), 3–5. <https://doi.org/10.1038/s43016-019-0010-4>
- Rockström, J., Thilsted, S. H., Willett, W. C., Gordon, L. J., Herrero, M., Hicks, C. C., Mason-D'Croz, D., Rao, N., Springmann, M., Wright, E. C., Agustina, R., Bajaj, S., Bunge, A. C., Carducci, B., Conti, C., Covic, N., Fanzo, J., Forouhi, N. G., Gibson, M. F., ...

- DeClerck, F. (2025). The EAT–Lancet Commission on healthy, sustainable, and just food systems. *The Lancet*, 406(10512), 1625–1700. [https://doi.org/10.1016/S0140-6736\(25\)01201-2](https://doi.org/10.1016/S0140-6736(25)01201-2)
- Ryan-Collins, J. (2019). *Beyond Voluntary Disclosure: Why a 'market-shaping' approach to financial regulation is needed to meet the challenge of climate change* (tech. rep.). UCL Insititute for Innovation and Public Purpose. Retrieved May 27, 2026, from <https://www.ucl.ac.uk/bartlett/sites/bartlett/files/iipp-pb-06-beyond-voluntary-disclosure-24-04-2019.pdf>
- S3F. (2026, April). Stop Financing Factory Farming | Join the Movement to End Factory Farming. Retrieved April 9, 2026, from <https://stopfinancingfactoryfarming.com>
- Salisu, A., Isah, K. O., & Ogbonna, A. E. (2025). Sectoral Corporate Profits and Long-Run Stock Return Volatility in the United States: A GARCH-MIDAS Approach. *Journal of Forecasting*, 44(2), 623–634. <https://doi.org/10.1002/for.3207>
- Searchinger, T., Waite, R., Hanson, C., & Ranganathan, J. (2019). *Creating a Sustainable Food Future: A Menu of Solutions to Feed Nearly 10 Billion People by 2050* (tech. rep.). World Resources Institute. Retrieved March 6, 2026, from <https://research.wri.org/wrr-food>
- Seferidi, P., Scrinis, G., Huybrechts, I., Woods, J., Vineis, P., & Millett, C. (2020). The neglected environmental impacts of ultra-processed foods. *The Lancet Planetary Health*, 4(10), e437–e438. [https://doi.org/10.1016/S2542-5196\(20\)30177-7](https://doi.org/10.1016/S2542-5196(20)30177-7)
- Semieniuk, G., Campiglio, E., Mercure, J.-F., Volz, U., & Edwards, N. R. (2021). Low-carbon transition risks for finance. *WIREs Climate Change*, 12(1), e678. <https://doi.org/10.1002/wcc.678>
- Semieniuk, G., Holden, P. B., Mercure, J.-F., Salas, P., Pollitt, H., Jobson, K., Vercoulen, P., Chewpreecha, U., Edwards, N. R., & Viñuales, J. E. (2022). Stranded fossil-fuel assets translate to major losses for investors in advanced economies. *Nature Climate Change*, 12(6), 532–538. <https://doi.org/10.1038/s41558-022-01356-y>
- Sharpe, W. F. (1966). Mutual Fund Performance. *The Journal of Business*, 39(1), 119–138. Retrieved April 2, 2026, from <https://www.jstor.org/stable/2351741>
- Shukla, P. R., Skea, J., Calvo Buendia, E., et al. (2019). *Climate change and land: An ipcc special report* (tech. rep.) (Stub entry: please verify full bibliographic details.). Intergovernmental Panel on Climate Change (IPCC).
- Silver, N. (2018). Blindness to risk: Why institutional investors ignore the risk of stranded assets. In *Stranded Assets*. Routledge.

- Sortino, F. A., & van der Meer, R. (1991). Downside Risk. *Journal of Portfolio Management*; 17(4). Retrieved April 2, 2026, from <https://www.proquest.com/openview/9698ae2e5a7f9005e71bb5a931a7ac96/1?cbl=49137&pq-origsite=gscholar>
- Stroebe, J., & Wurgler, J. (2021). What do you think about climate finance? *Journal of Financial Economics*, 142(2), 487–498. <https://doi.org/10.1016/j.jfineco.2021.08.004>
- te Wierik, S., DeClerck, F., Beusen, A., Gerten, D., Maggi, F., Norberg, A., Noone, K., Schulte-Uebbing, L., Springmann, M., Tang, F. H. M., de Vries, W., van Vuuren, D., Vermeulen, S., & Rockström, J. (2025). Identifying the safe operating space for food systems. *Nature Food*, 6(12), 1153–1163. <https://doi.org/10.1038/s43016-025-01252-6>
- Thomä, J., & Chenet, H. (2017). Transition risks and market failure: A theoretical discourse on why financial models and economic agents may misprice risk related to the transition to a low-carbon economy. *Journal of Sustainable Finance & Investment*, 7(1), 82–98. <https://doi.org/10.1080/20430795.2016.1204847>
- Trinks, A., Scholtens, B., Mulder, M., & Dam, L. (2018). Fossil Fuel Divestment and Portfolio Performance. *Ecological Economics*, 146, 740–748. <https://doi.org/10.1016/j.ecolecon.2017.11.036>
- Trouvé, B., & Naef, A. (2025, January). *Divestment From Fossil Fuels: Evidence From Ownership Data*. Retrieved April 8, 2026, from <https://essec.hal.science/hal-04918263> working paper or preprint.
- UNFCCC. (2025). *Baku to Belém Roadmap to 1.3T* (tech. rep.). Retrieved April 8, 2026, from <https://unfccc.int/topics/climate-finance/workstreams/baku-to-belem-roadmap-to-13t>
- van der Ploeg, F., & Rezai, A. (2020). Stranded Assets in the Transition to a Carbon-Free Economy. *Annual Review of Resource Economics*, 12(Volume 12, 2020), 281–298. <https://doi.org/10.1146/annurev-resource-110519-040938>
- Vermeulen, S. J., Campbell, B. M., & Ingram, J. S. I. (2012). Climate Change and Food Systems. *Annual Review of Environment and Resources*, 37(Volume 37, 2012), 195–222. <https://doi.org/10.1146/annurev-environ-020411-130608>
- Walton, S., Mehrabi, Z., Fanzo, J., & Caldecott, B. (2025). Asset stranding could open new pathways to food systems transformation. *Nature Food*, 6(5), 440–445. <https://doi.org/10.1038/s43016-025-01170-7>
- Webb, P., Benton, T. G., Beddington, J., Flynn, D., Kelly, N. M., & Thomas, S. M. (2020). The urgency of food system transformation is now irrefutable. *Nature Food*, 1(10), 584–585. <https://doi.org/10.1038/s43016-020-00161-0>
- Willett, W., Rockström, J., Loken, B., Springmann, M., Lang, T., Vermeulen, S., Garnett, T., Tilman, D., DeClerck, F., Wood, A., Jonell, M., Clark, M., Gordon, L. J., Fanzo, J.,

Hawkes, C., Zurayk, R., Rivera, J. A., Vries, W. D., Sibanda, L. M., . . . Murray, C. J. L. (2019). Food in the Anthropocene: The EAT–Lancet Commission on healthy diets from sustainable food systems. *The Lancet*, 393(10170), 447–492. [https://doi.org/10.1016/S0140-6736\(18\)31788-4](https://doi.org/10.1016/S0140-6736(18)31788-4)

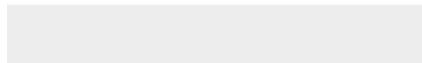
WWF. (2022). *Seeing the Forest for the Trees: A Practical Guide for Financial Institutions to take Action against Deforestation and Conversion Risks* (tech. rep.). Retrieved March 6, 2026, from https://wwfint.awsassets.panda.org/downloads/seeing_the_forest_for_the_trees.pdf

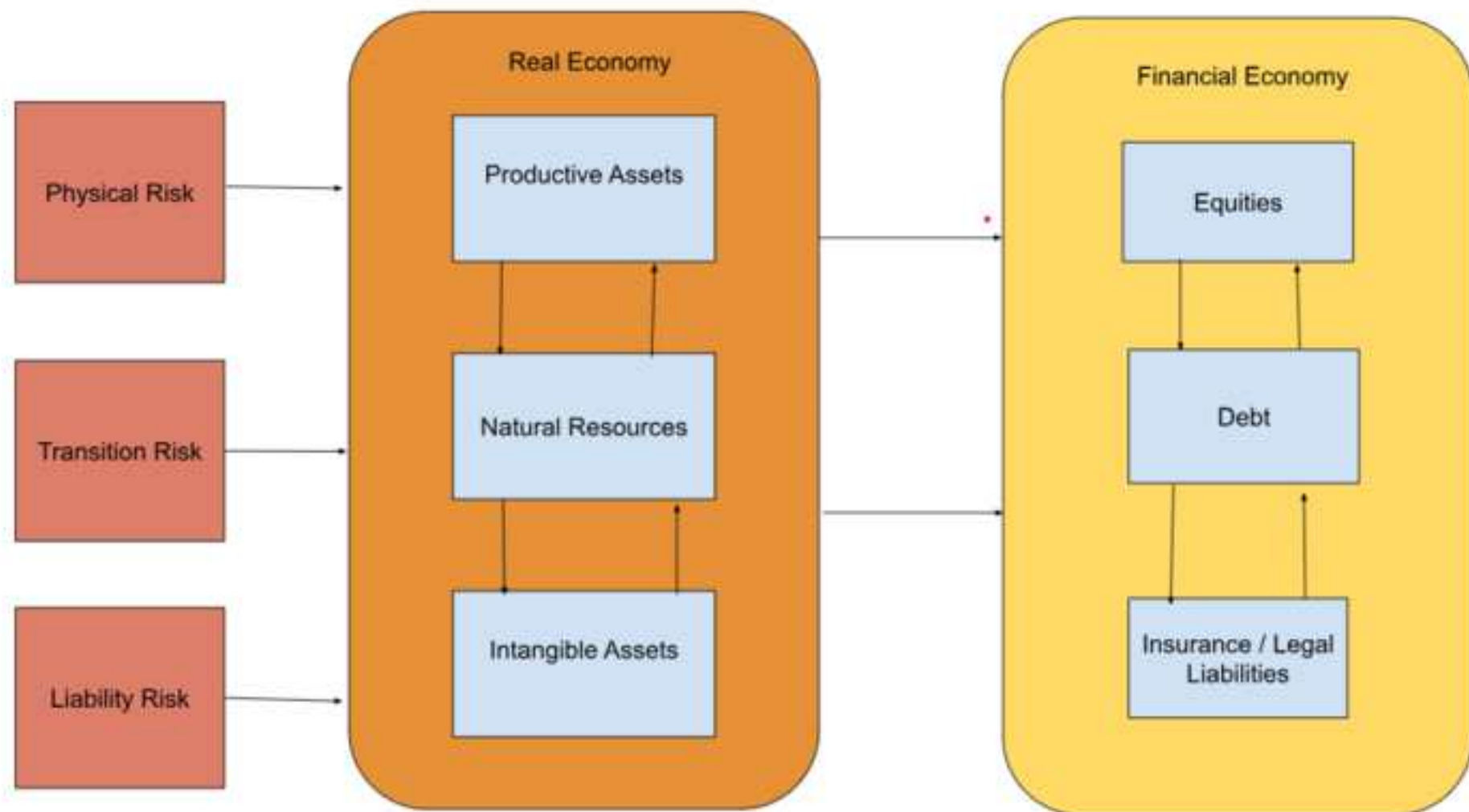


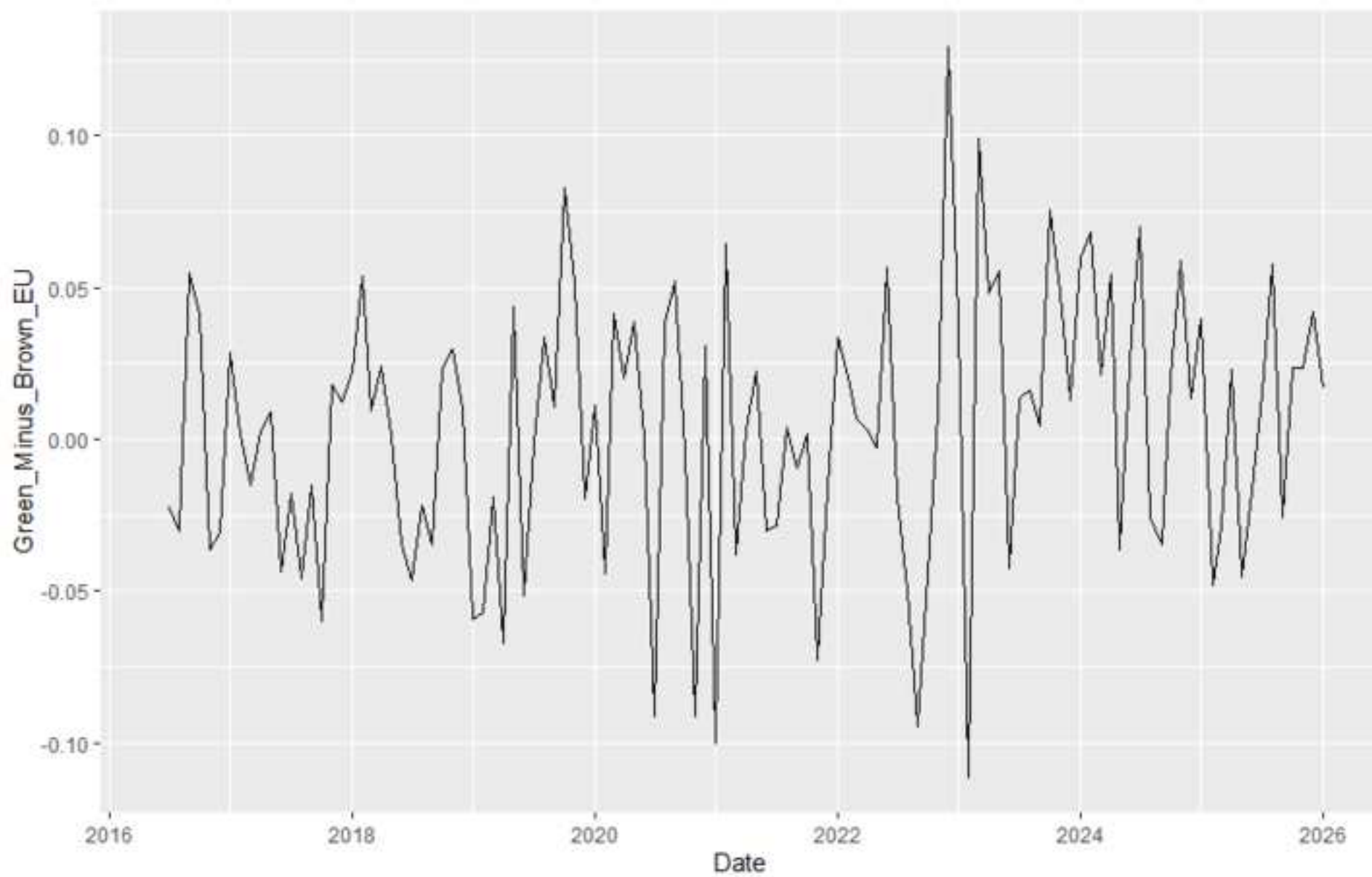
[Click here to access/download](#)

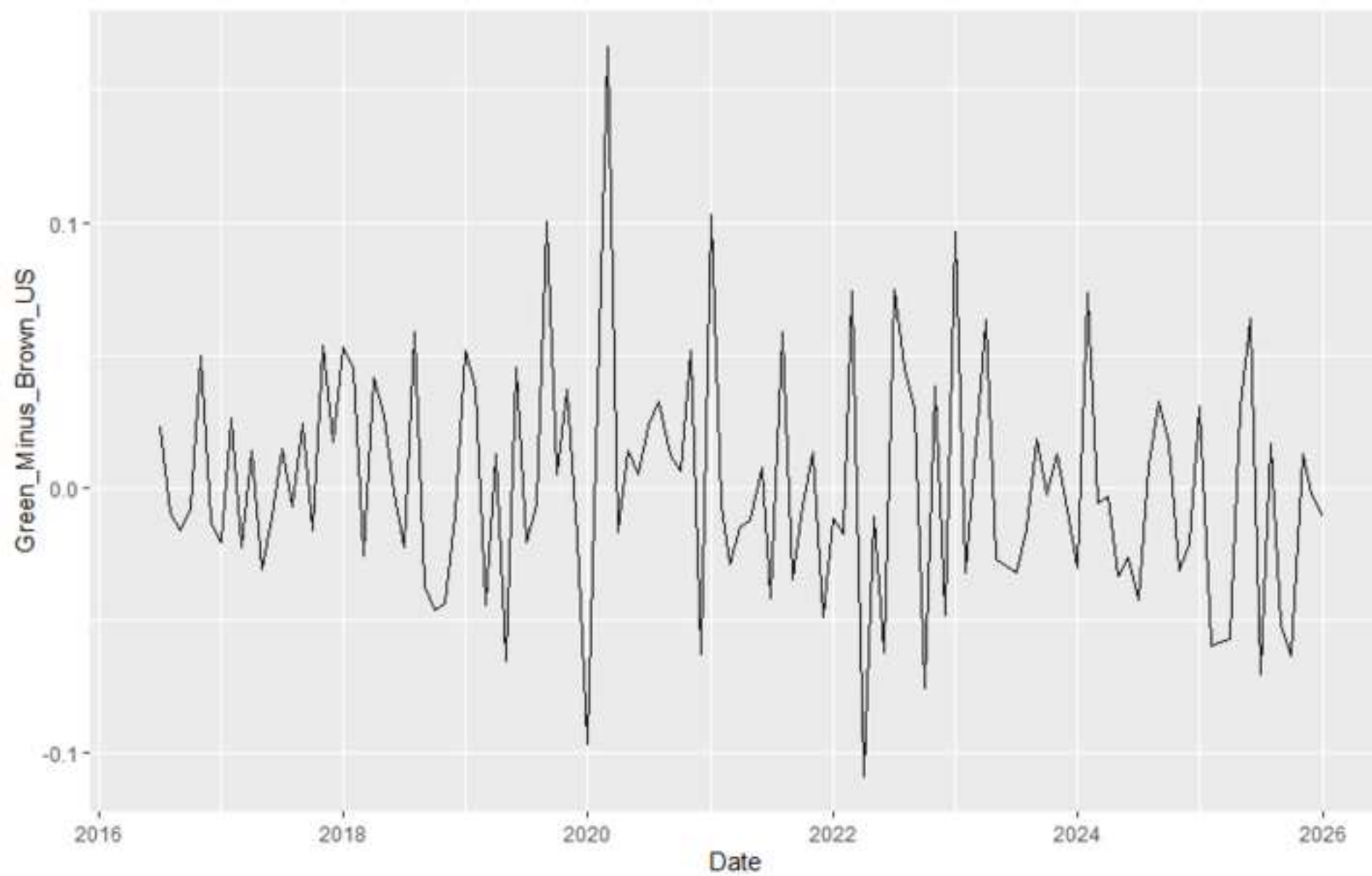
LaTeX source files

NFG_P1_Manuscript_Source_File.tex









Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: